Class Manager

Group 8

MILESTONE 2

Simão Tojal 119636
Gonçalo Almeida 119792
Margarida Ribeiro 119876
João Barreira 120054
José Bagagem 120141
Professor João Almeida e Daniel Ferreira









ClassManager



“

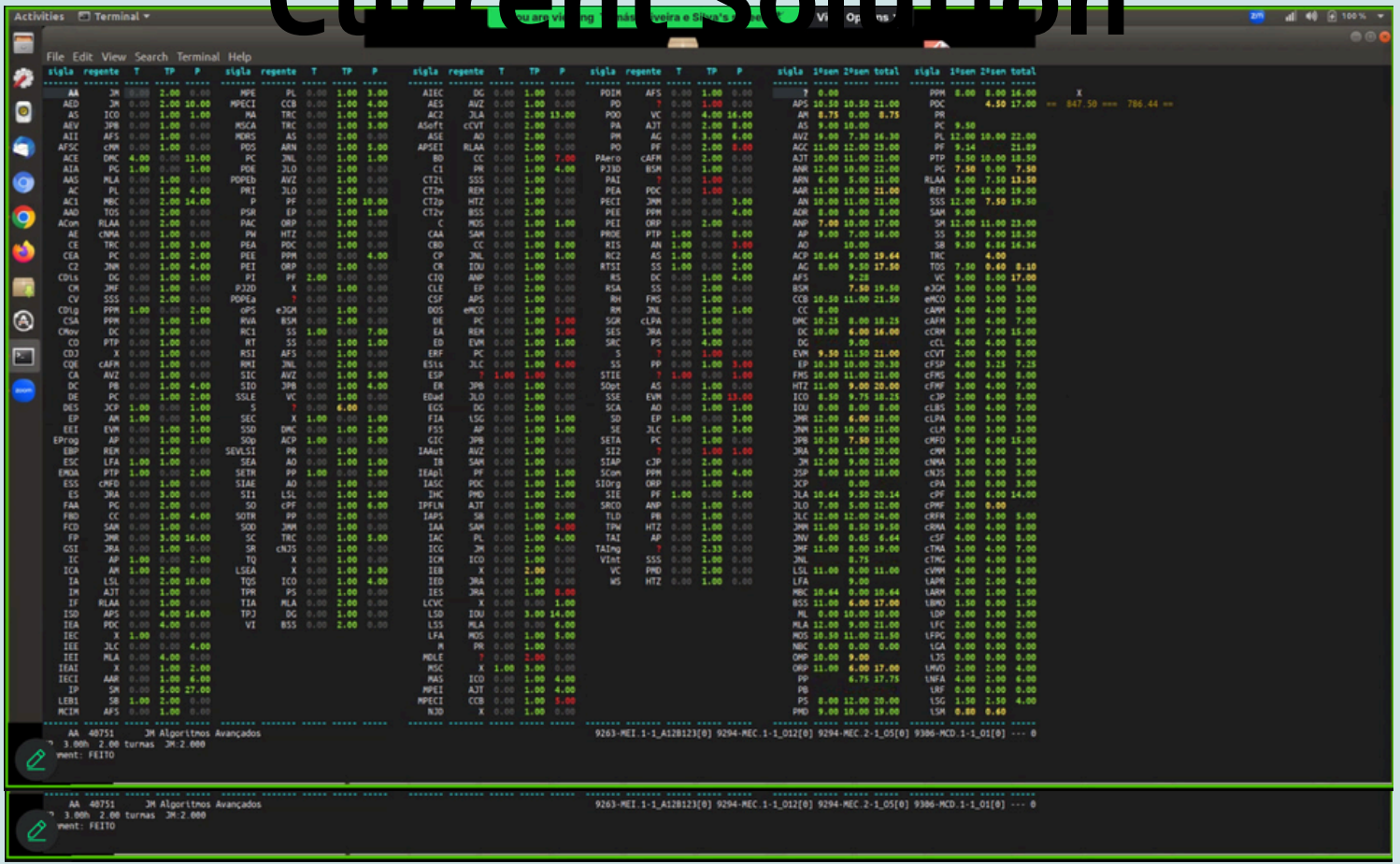
is a system to manage the distribution of schedules for teachers, classrooms, and students.

”

Current Solution

Room	Class	Time	Professor
Class: 4.2.1	IA P3	Time: 14h	Professor: Júao
Class: 4.2.8	IA P2	Time: 9h	Professor: Tiago
Class: Anv 5	SIO T1	Time: 11h	Professor: Margarida
Class: Anv 4	IA T1	Time: 15h	Professor: Tojal
Class: 4.2.10	FP P1	Time: 17h	Professor: Almeida
Class: 4.1.5	SO P4	Time: 10h	Professor: Barreira
Class: 4.3.8	RS 08	Time: 11h	Professor: Bagagem
Class: 4.1.108	AED P1	Time: 13h	Professor: Almeida

Current Solution



Personas

Receives notifications about schedule/room changes.



Student



Secretary

User with limited IT experience, manages schedules, rooms, notifications.

Validates their schedule, declares availability, checks room conditions.

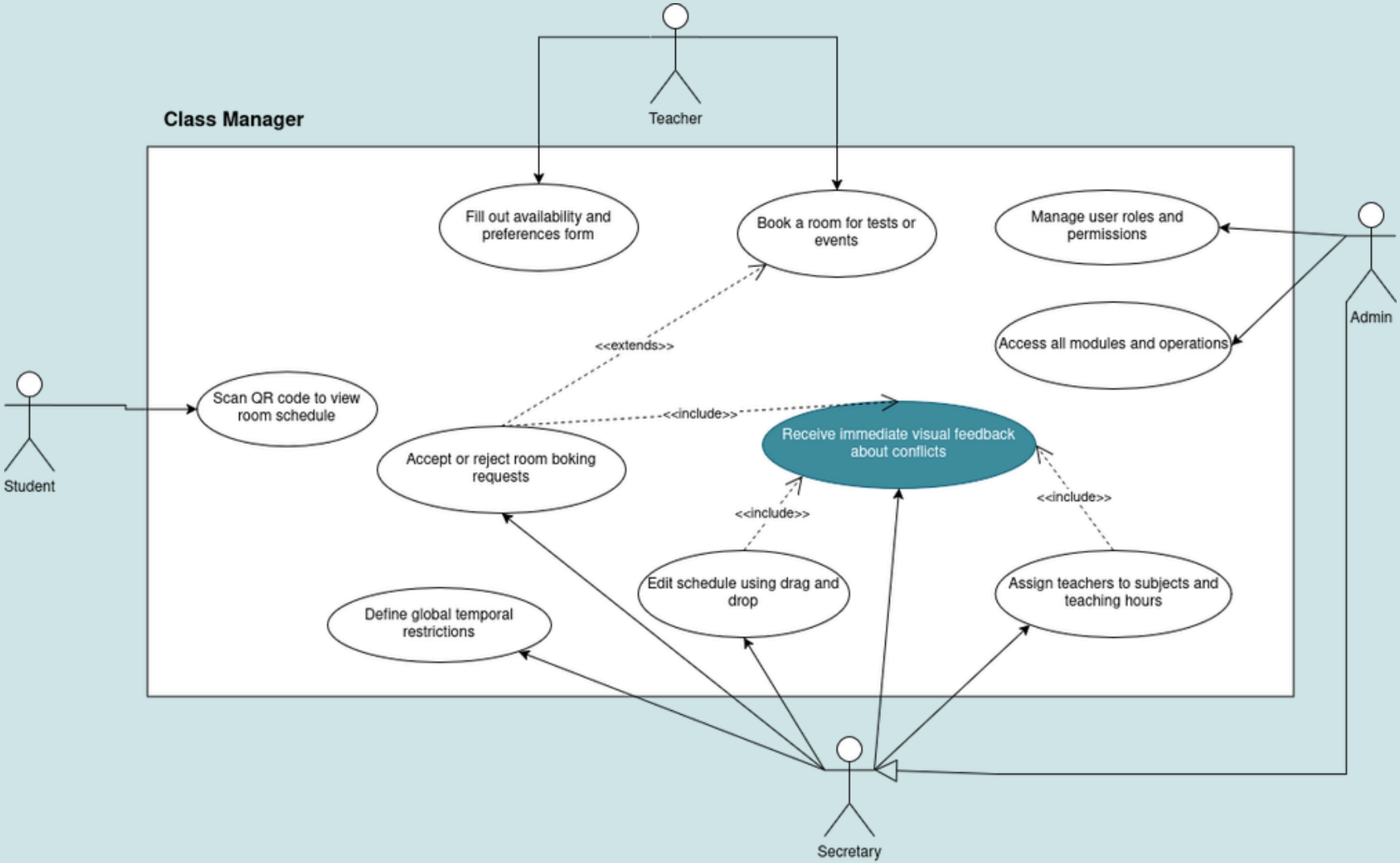


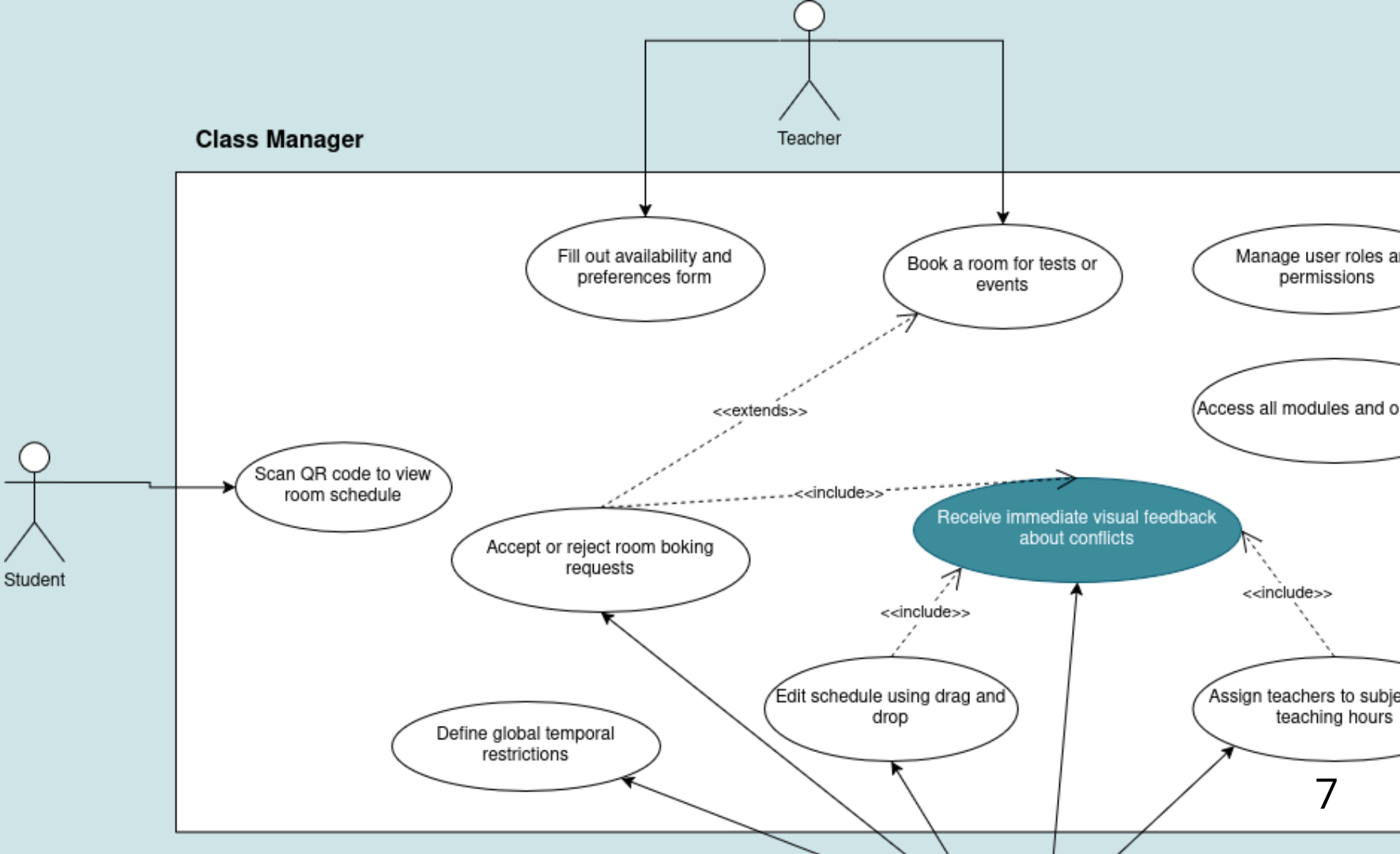
Teacher

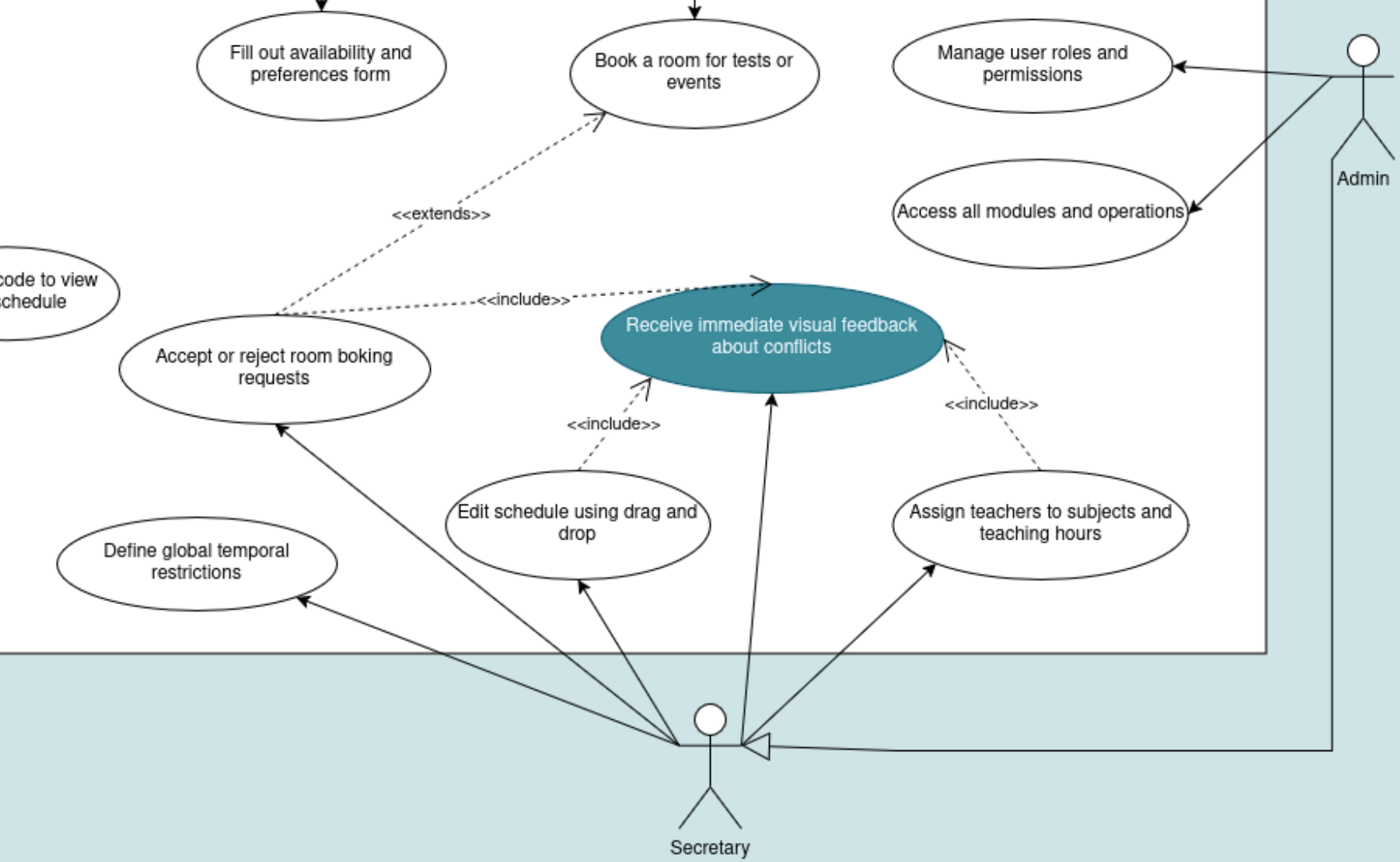


Admin

Super user, full privileges, supports Secretary.


Class Manager
Teacher
Fill out availability and preferences form
Book a room for tests or events
Manage user roles and permissions
Admin
Access all modules and operations
<<extends>>
Student
Scan QR code to view room schedule
<<include>>
Receive immediate visual feedback about conflicts
Accept or reject room boking requests
<<include>>
<<include>>
Edit schedule using drag and drop
Assign teachers to subjects and teaching hours
Define global temporal restrictions
Secretary


Class Manager
Teacher
Fill out availability and preferences form
Book a room for tests or events
Manage user roles a permissions
Access all modules and o
<<extends>>
Student
Scan QR code to view room schedule
<<include>>
Accept or reject room boking requests
Receive immediate visual feedback about conflicts
<<include>>
<<include>>
Edit schedule using drag and drop
Assign teachers to subje teaching hours
Define global temporal restrictions


Fill out availability and preferences form
Book a room for tests or events
Manage user roles and permissions
Admin
Access all modules and operations
<<extends>>
code to view schedule
<<include>>
Receive immediate visual feedback about conflicts
Accept or reject room boking requests
<<include>>
<<include>>
Edit schedule using drag and drop
Assign teachers to subjects and teaching hours
Define global temporal restrictions
Secretary

Functional Requirements

FR1 The system must allow authorized users to **add**, **edit**, and **remove academic entities**, including (but not limited to) subjects (UCs), teachers, groups/classes, and classrooms.

FR2 The system must allow for **association between teachers and subjects** and **define teaching hours per class**.

FR3 The system must allow **setting global academic periods** and **specific unavailability for schedules.**

FR4 The system must send **automatic notifications** when **changes** occur in **schedules** or **rooms.**

Functional Requirements

FR5 The system must allow **teachers** to **submit availability and preferences** to influence **schedule planning**.

FR6 The system must allow **teachers** to **request room reservations** and **receive approval or rejection** feedback.

FR7 The system must allow **students** to **scan** a **room's QR code** to instantly **view its timetable.**

FR8 The system must **provide suggestions** to **resolve scheduling conflicts.**

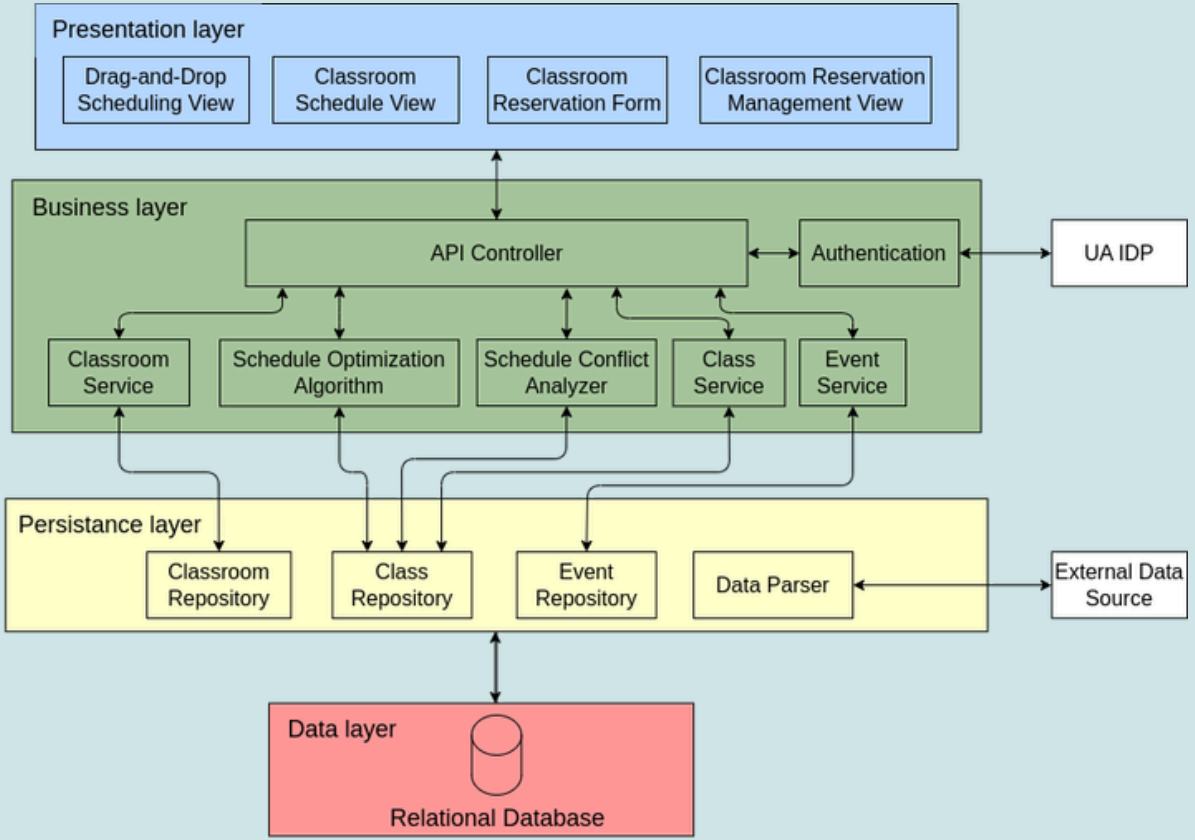
Non-Functional Requirements

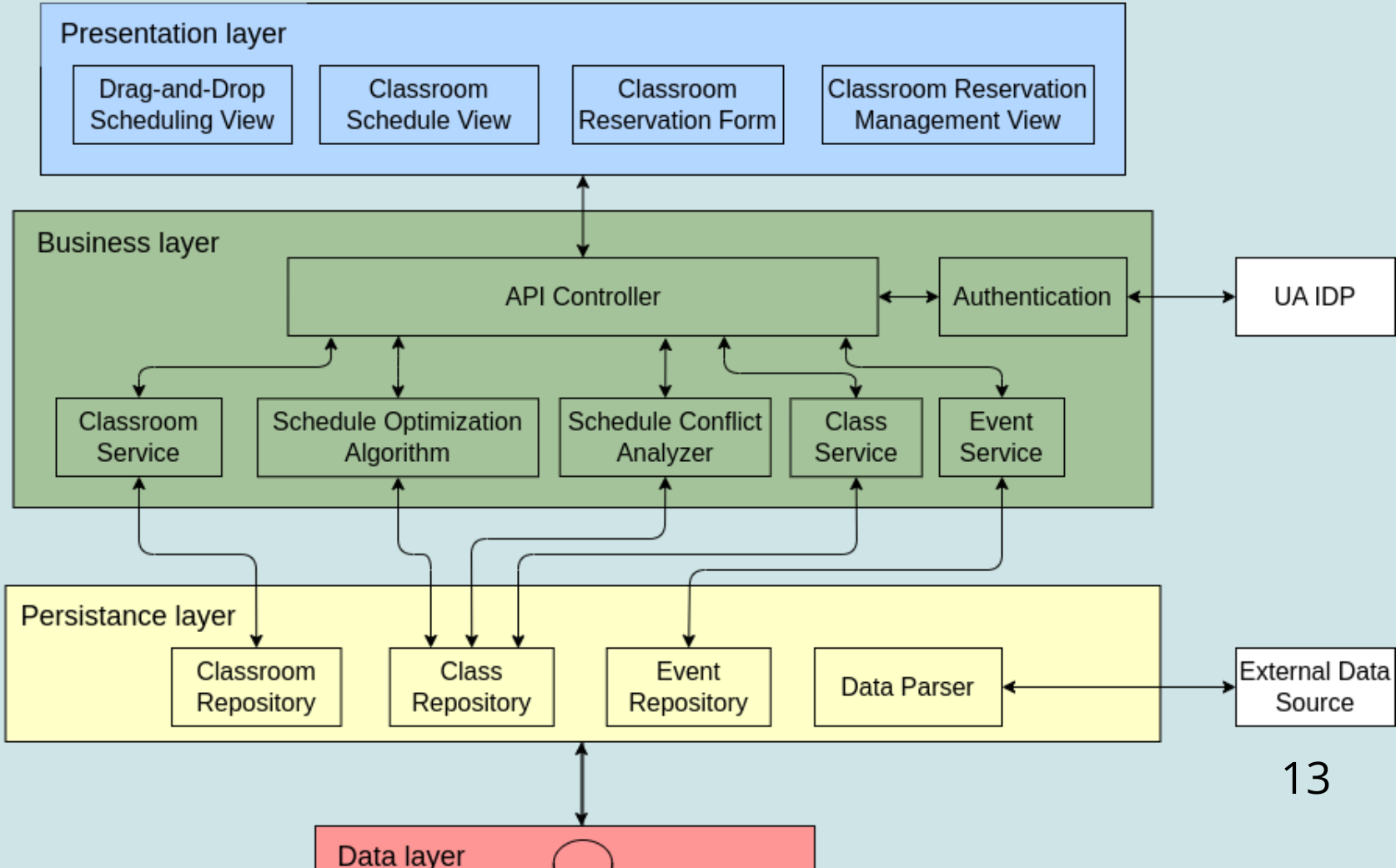
NFR1 The system should integrate the **University of Aveiro's Idp**, allowing **users** to **securely log** in using their **institutional credentials.**

NFR2 The **interface** should be **intuitive**, **visually clear**, and **easy to use** for secretaries, teachers, and students through the **following** of **Nielsen's 10 Usability Heuristics.**

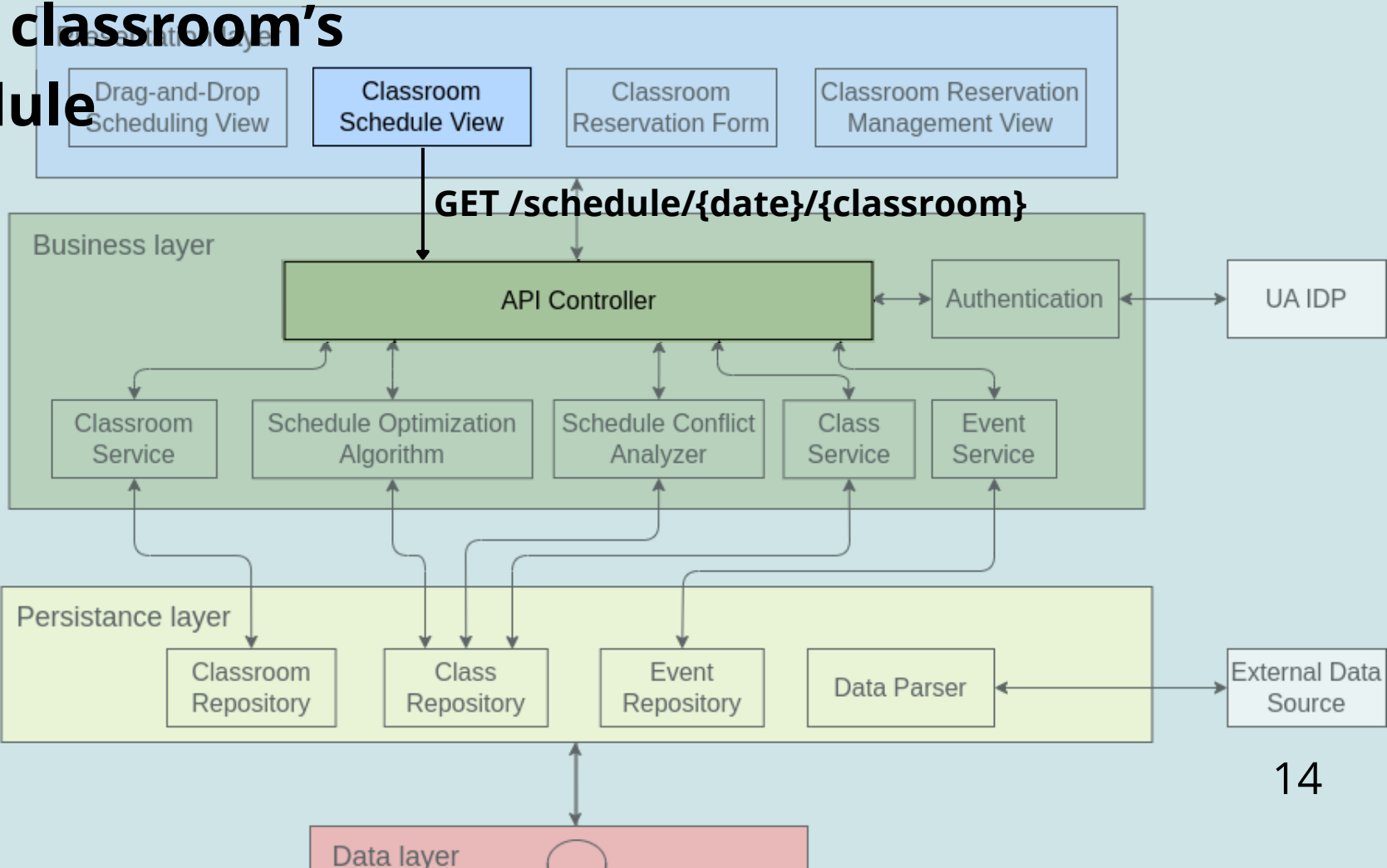
NFR3 All **API** endpoints should be **protected** against **SQL injection**, **XSS** and **CSRF attacks** through tools like **OWASP ZAP.**

Architecture

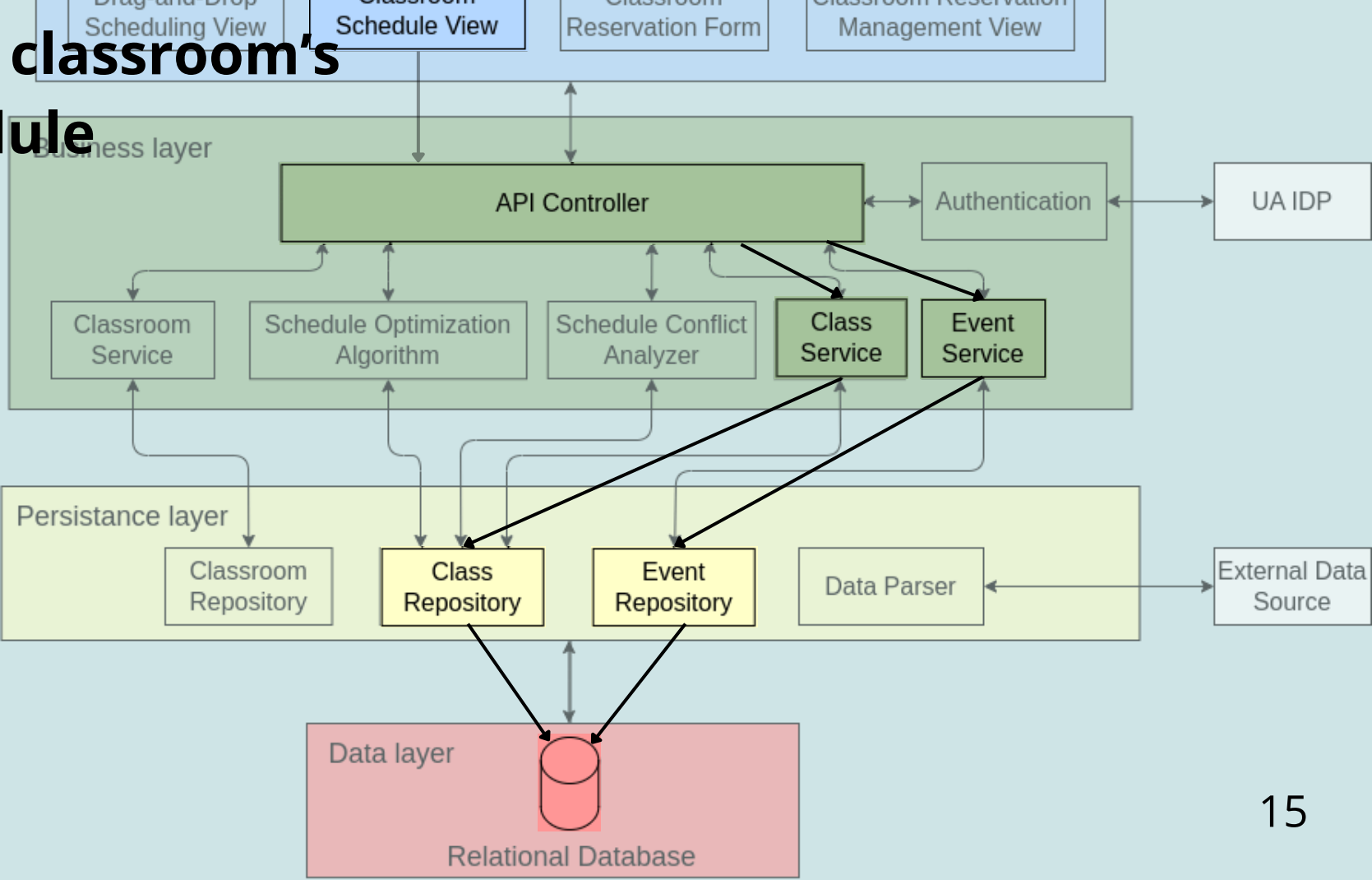



Presentation layer
Drag-and-Drop Scheduling View
Classroom Schedule View
Classroom Reservation Form
Classroom Reservation Management View
Business layer
API Controller
Authentication
UA IDP
Classroom Service
Schedule Optimization Algorithm
Schedule Conflict Analyzer
Class Service
Event Service
Persistance layer
Classroom Repository
Class Repository
Event Repository
Data Parser
External Data Source
Data layer

Show classroom's schedule

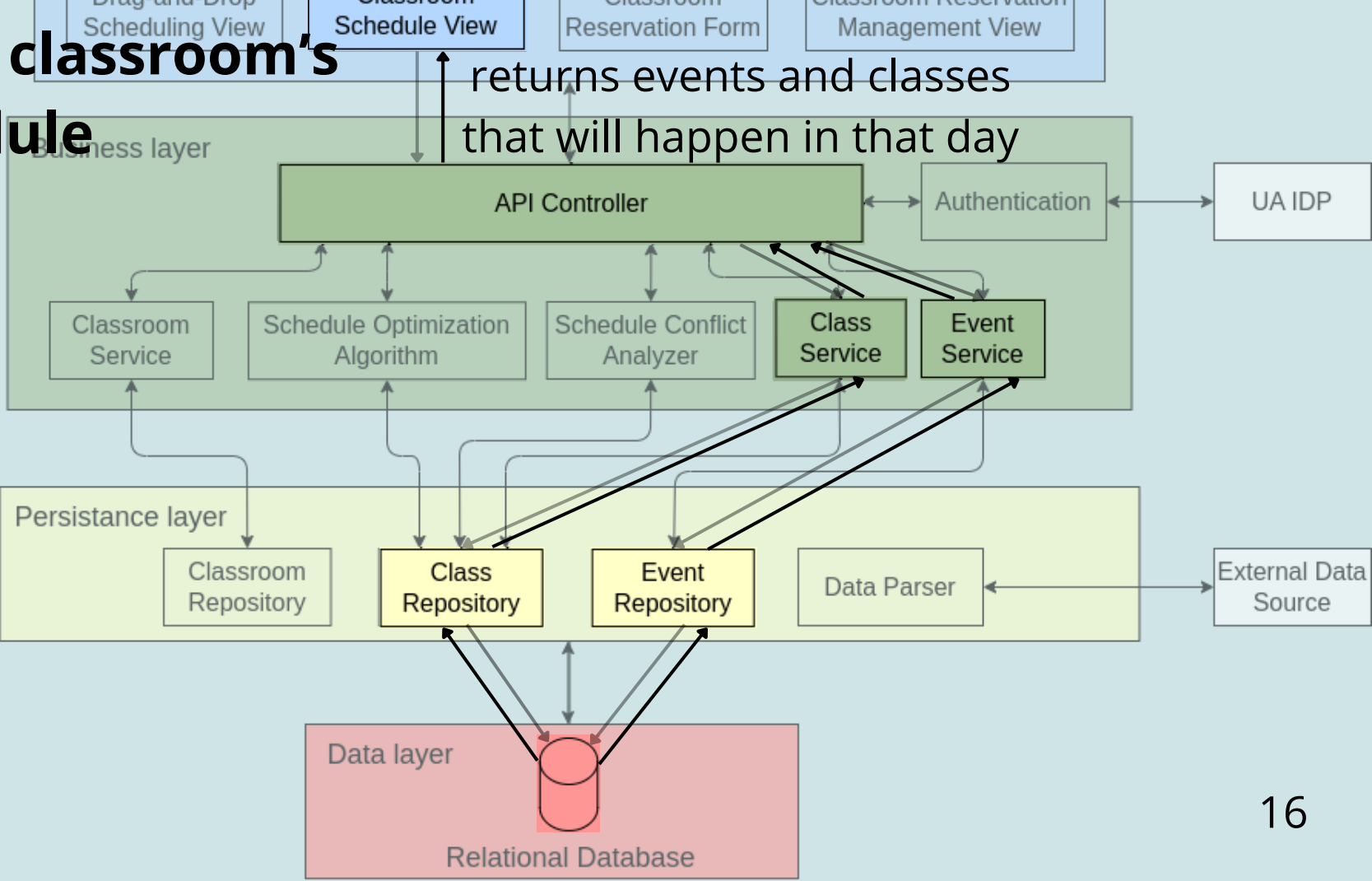


Show classroom's schedule

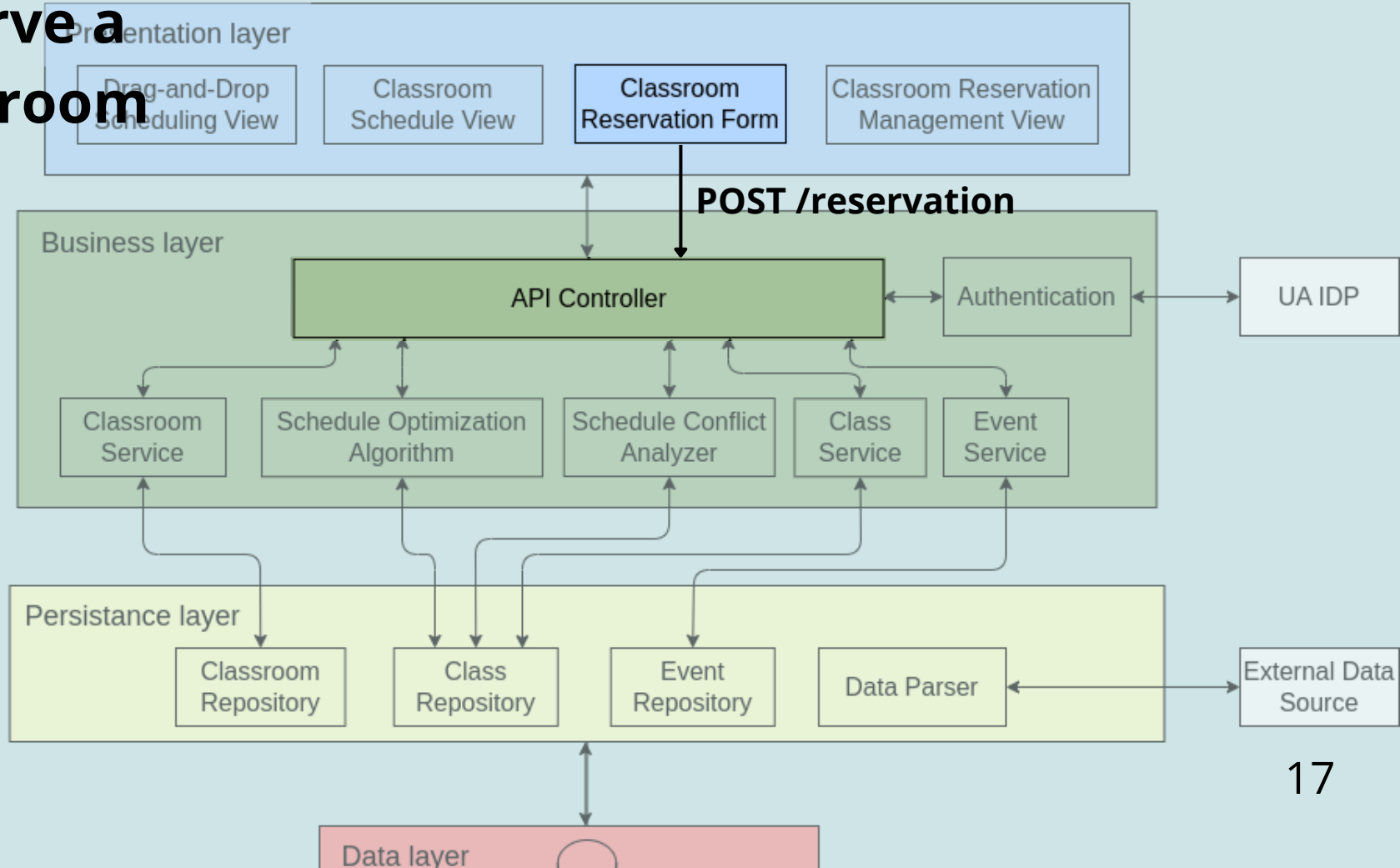


Show classroom's schedule

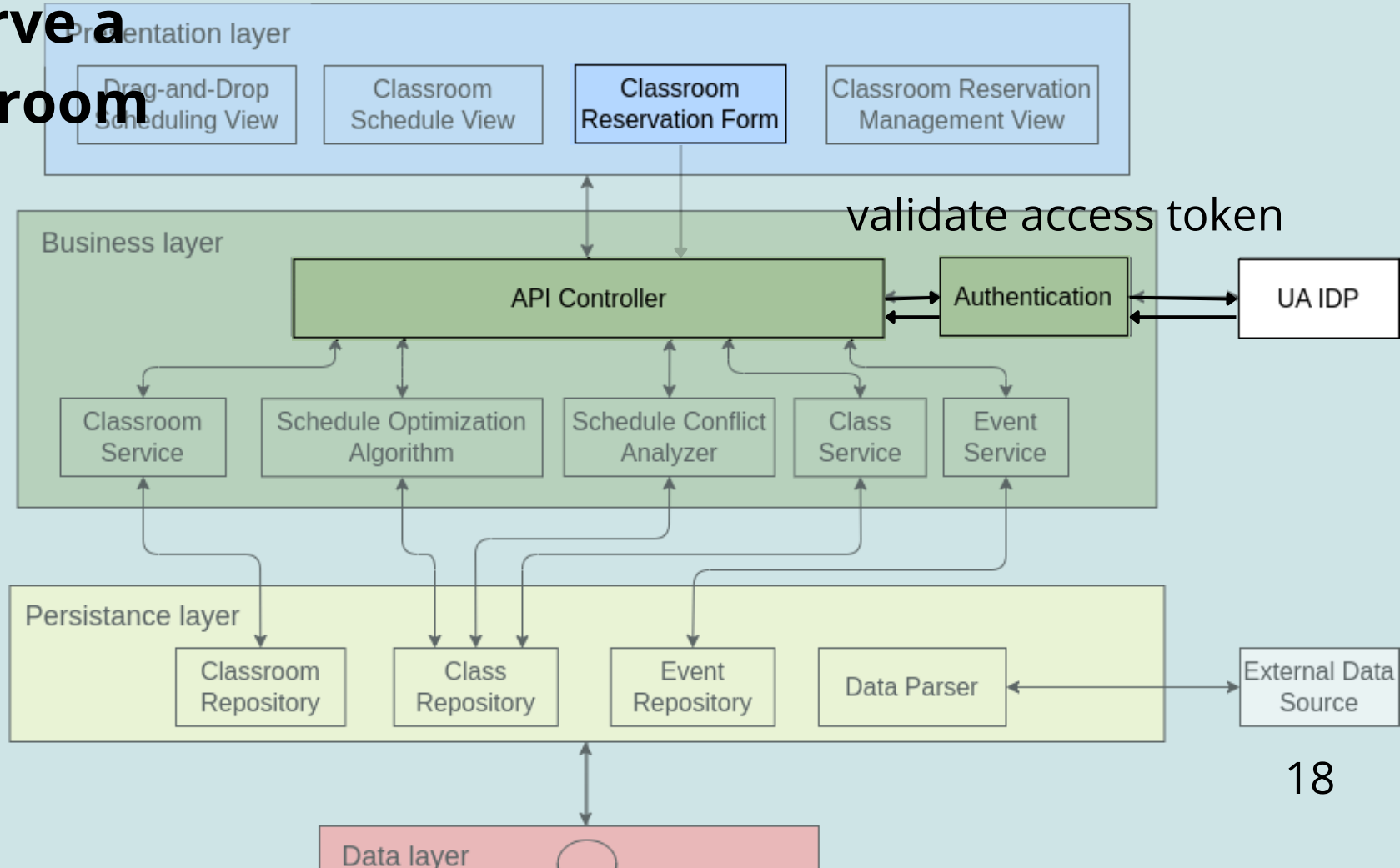
returns events and classes that will happen in that day



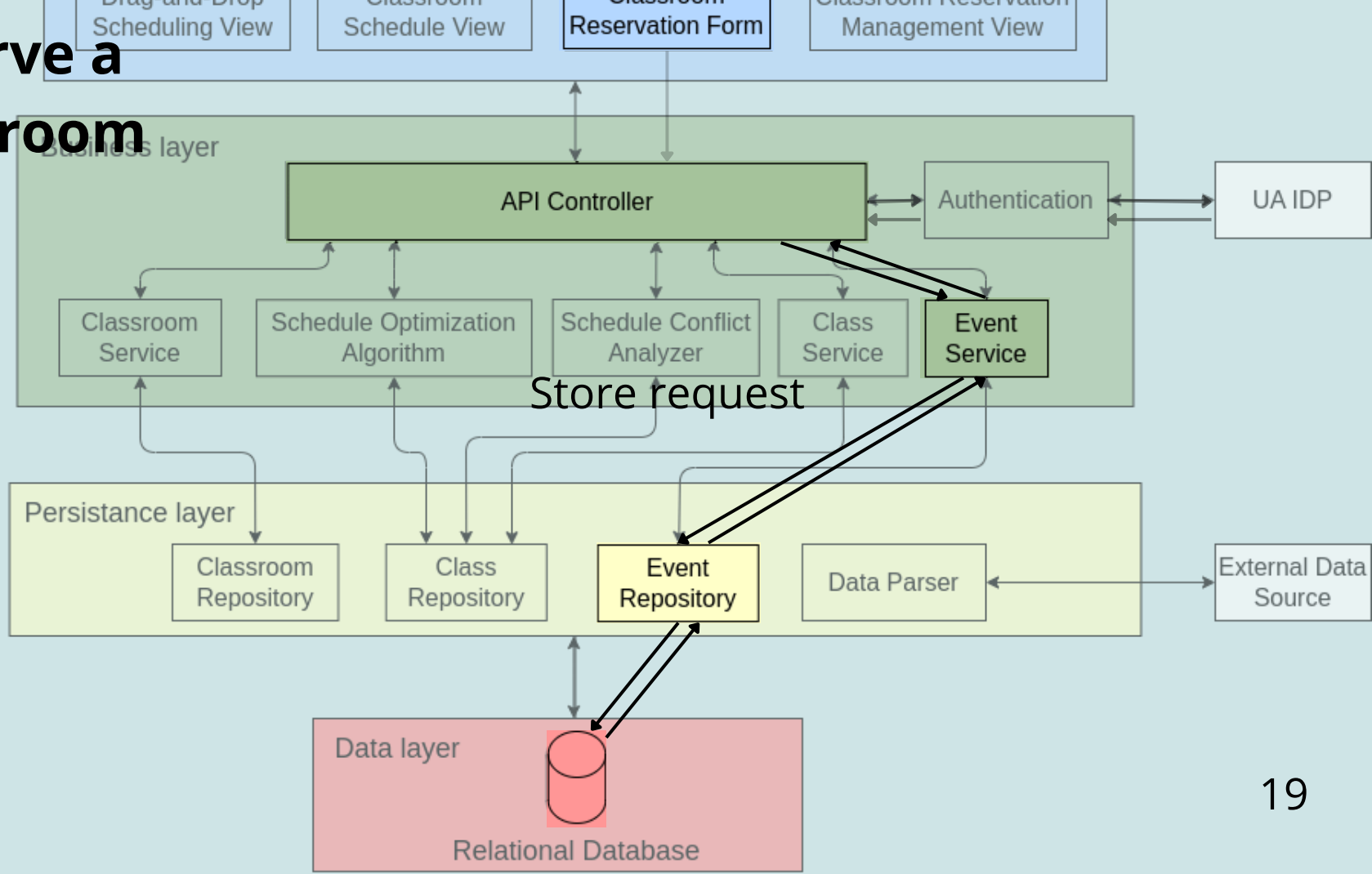
Reserve a Classroom



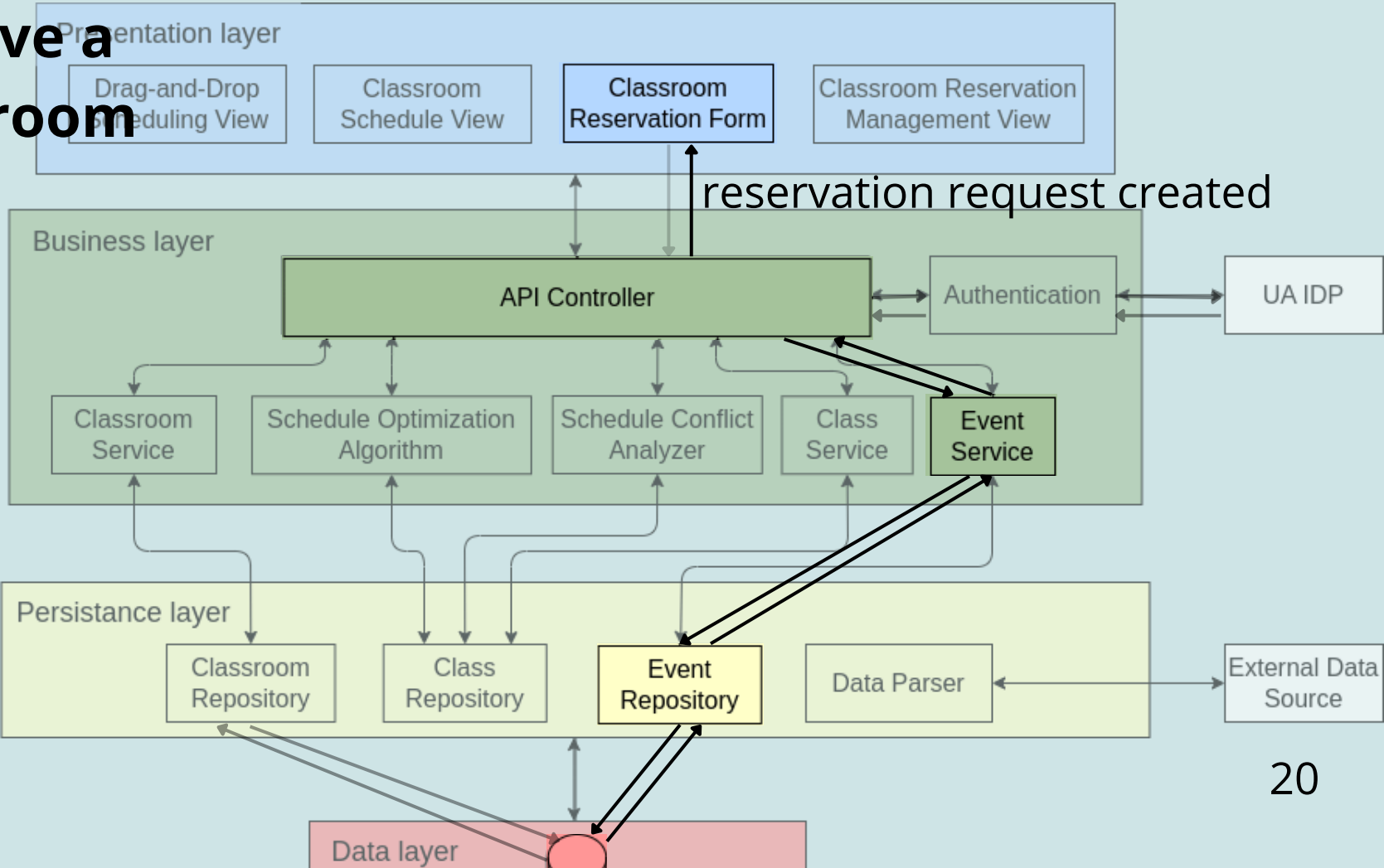
Reserve a Classroom



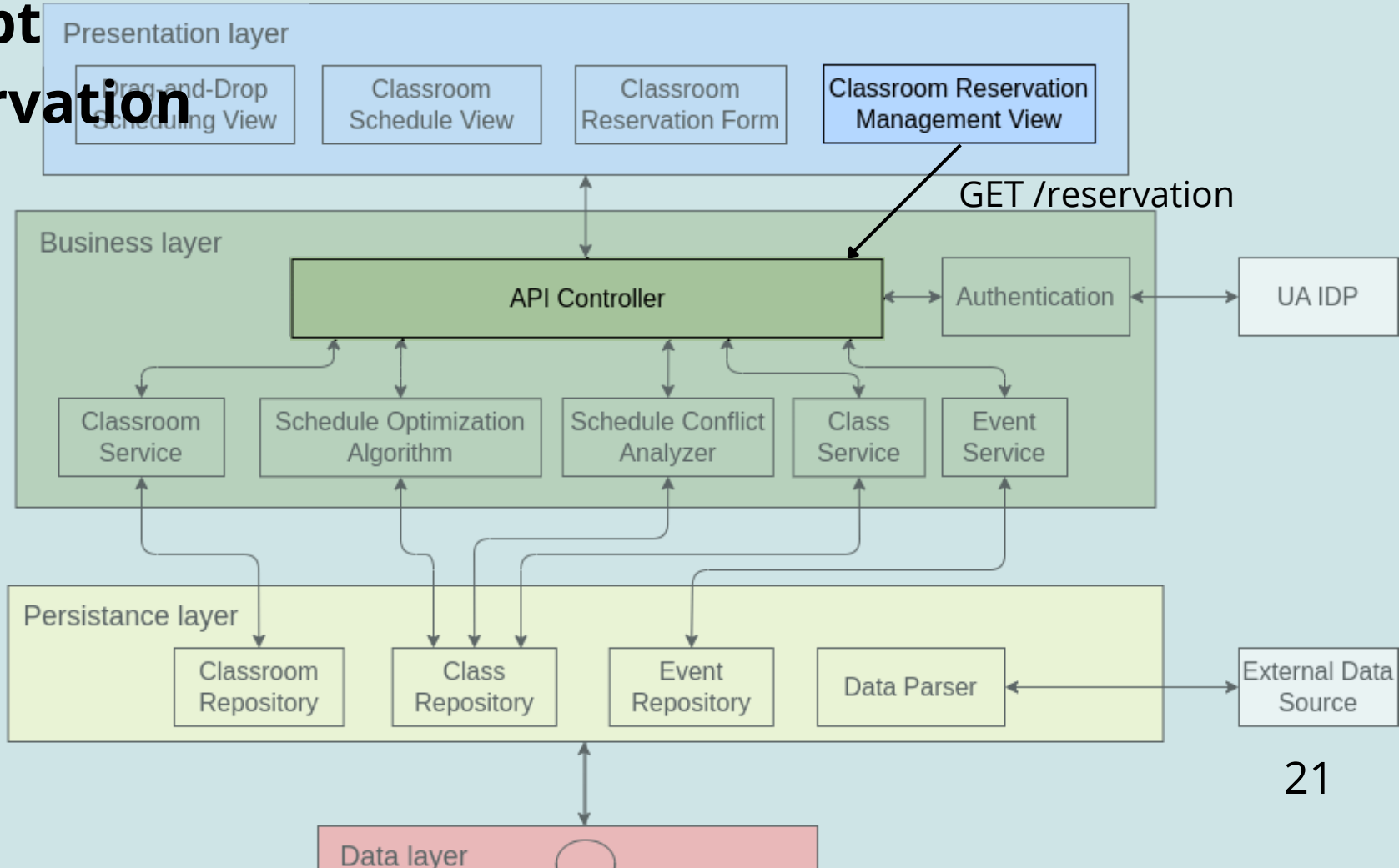
Reserve a Classroom



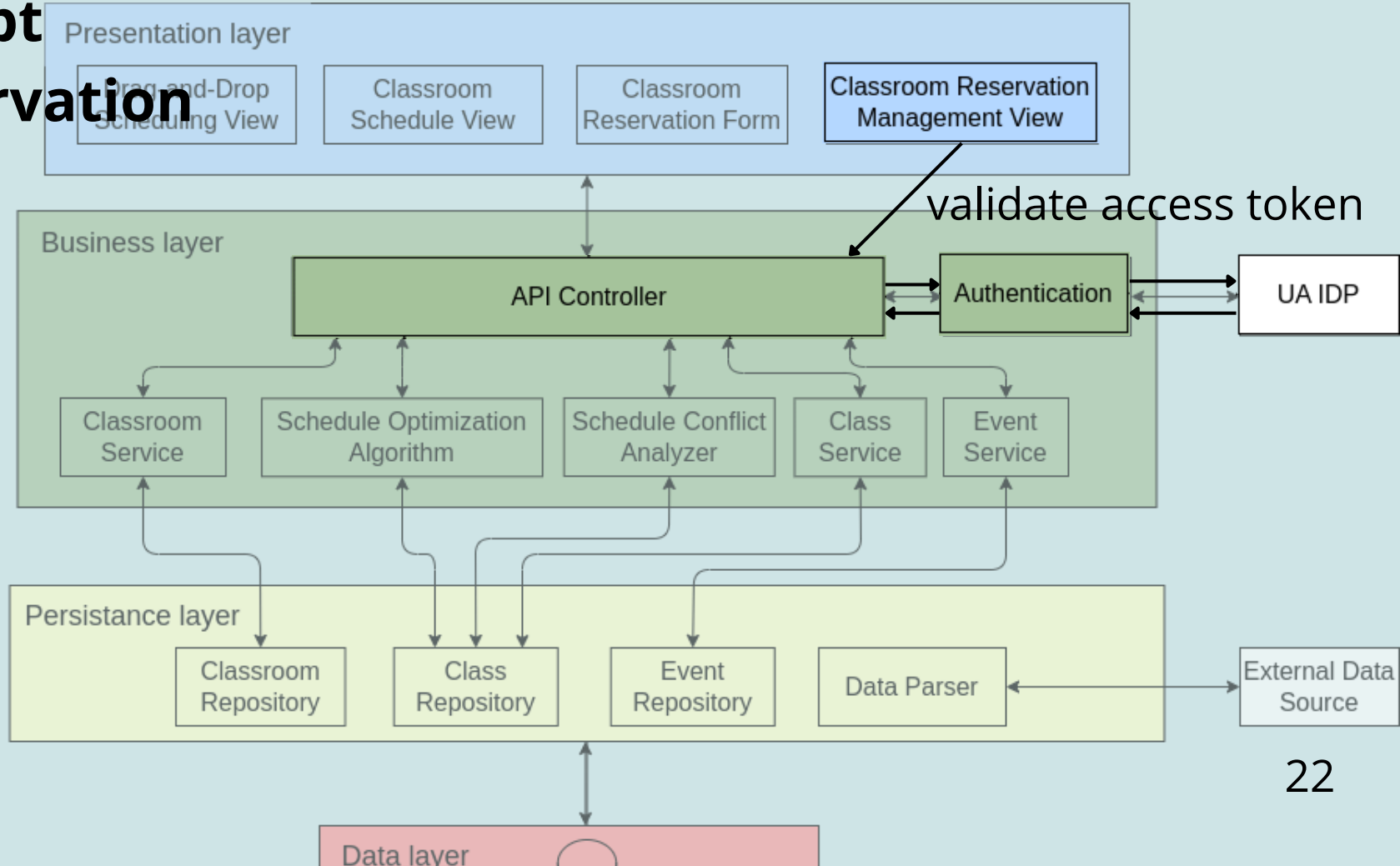
Reserve a Classroom



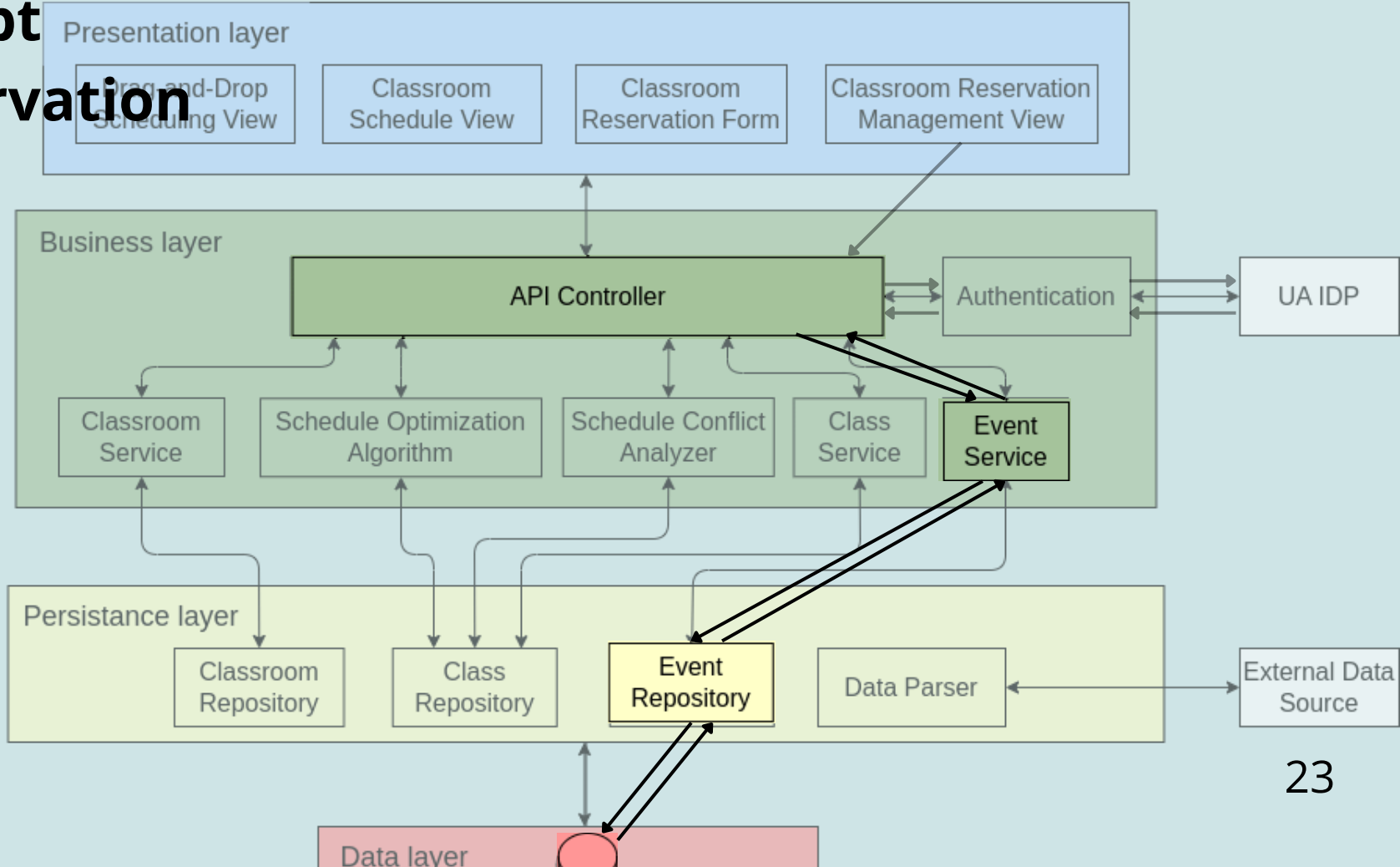
Accept Reservation



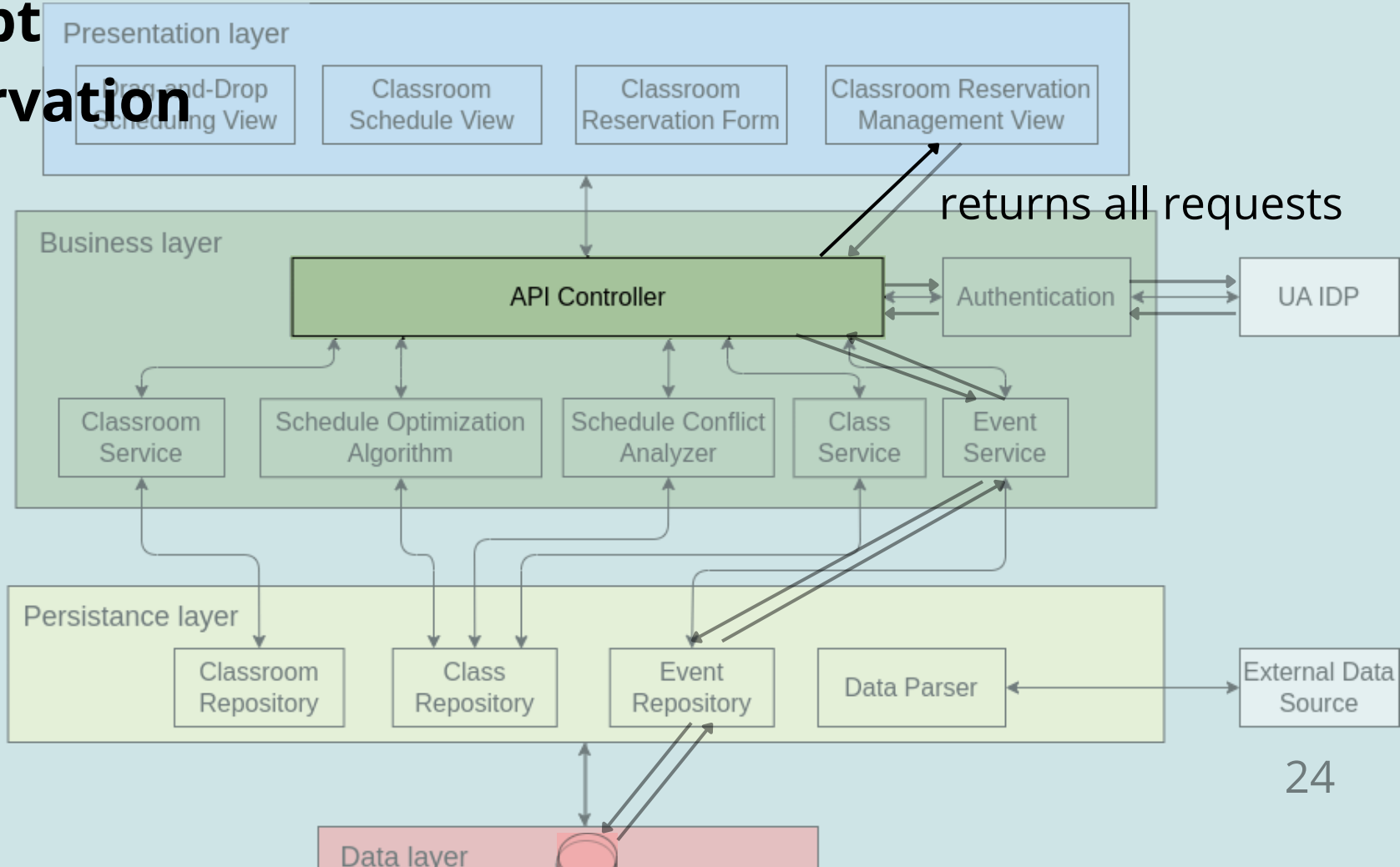
Accept Reservation



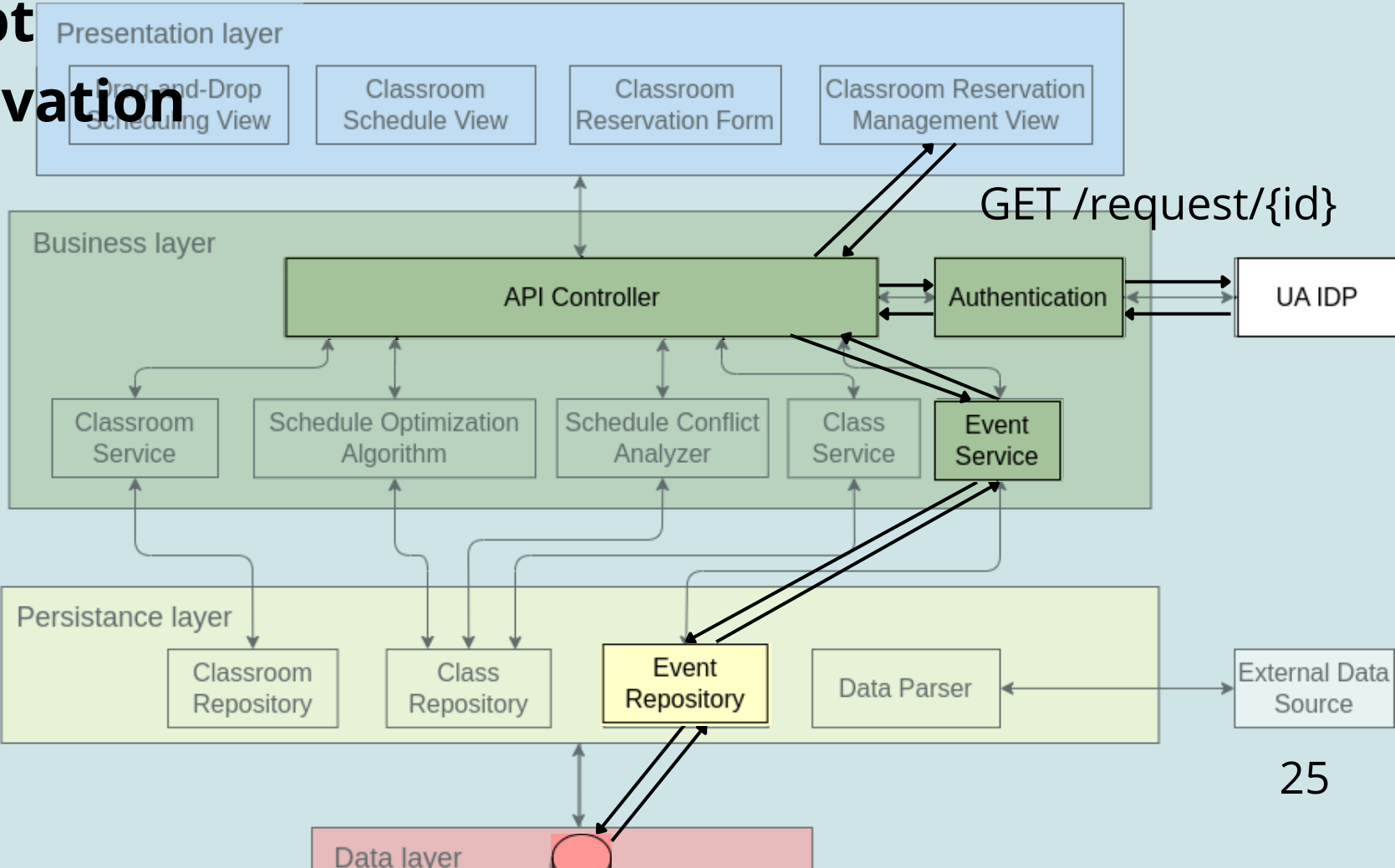
Accept Reservation



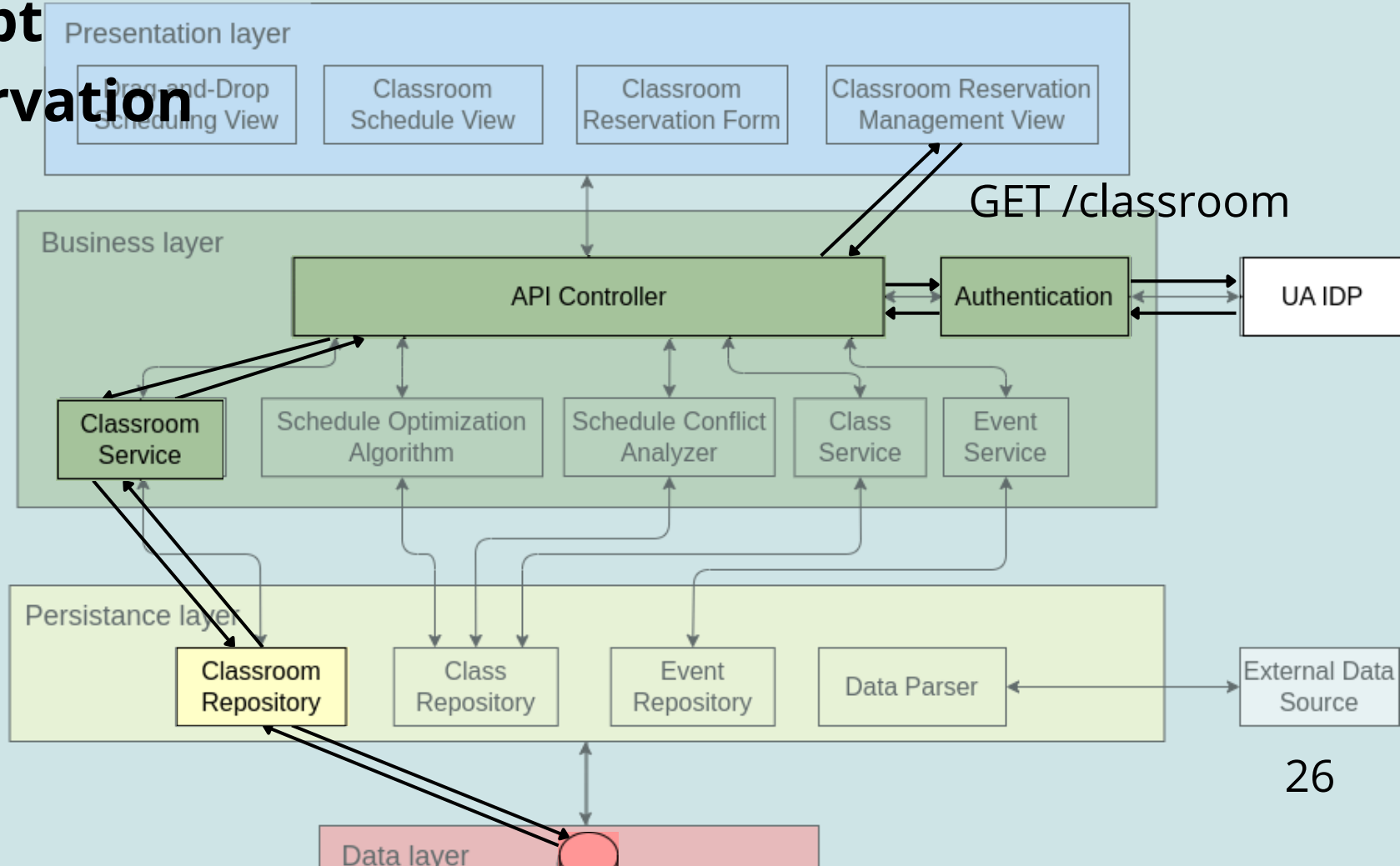
Accept Reservation



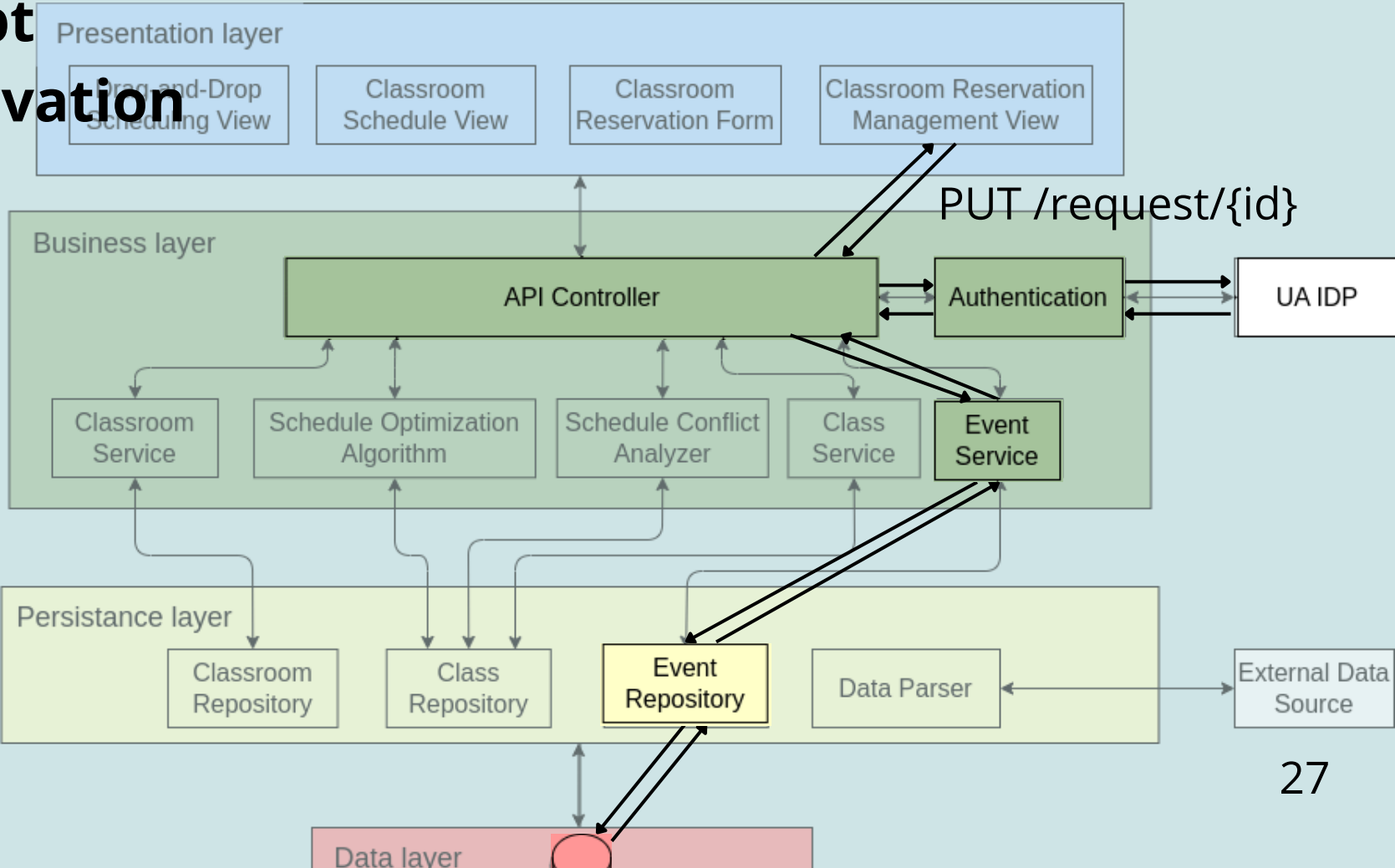
Accept Reservation



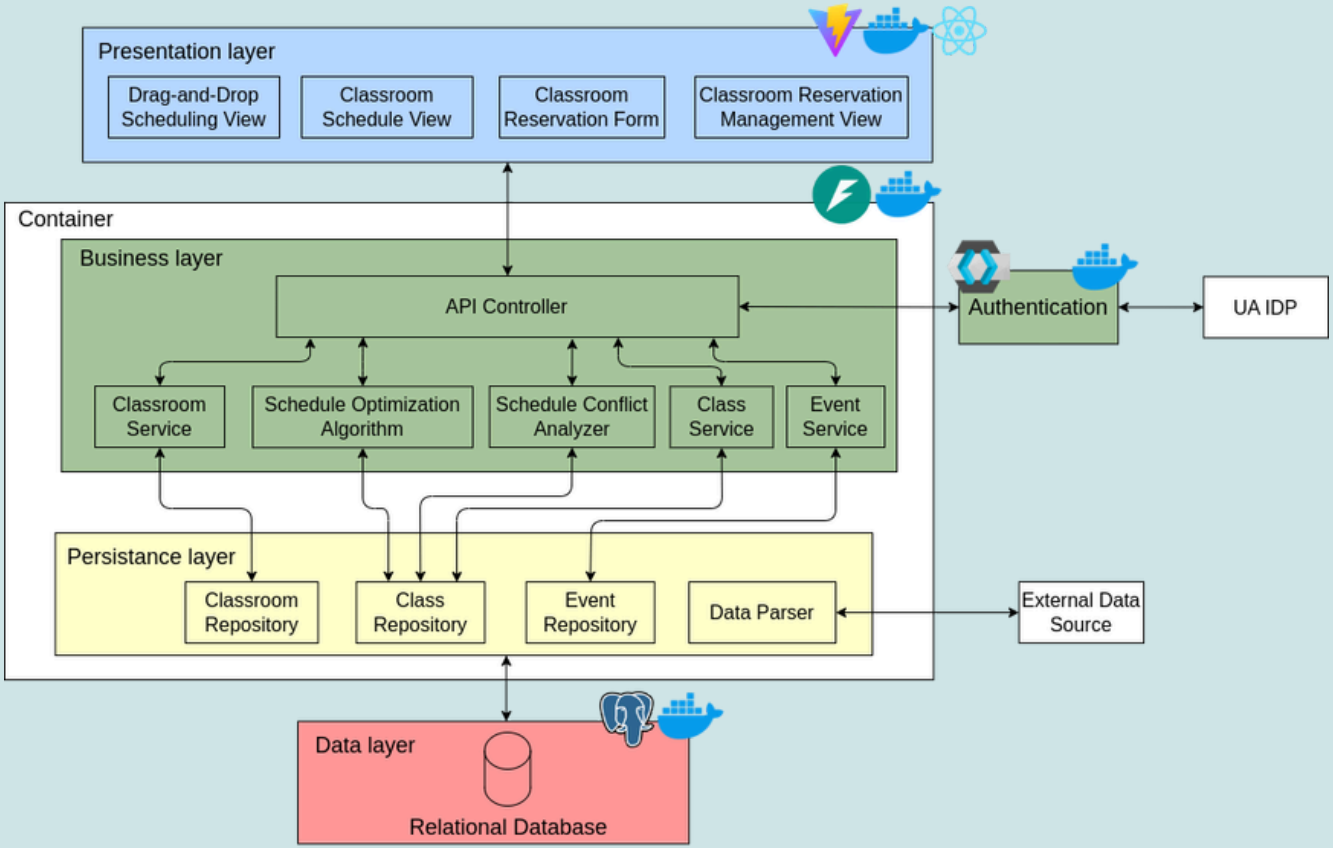
Accept Reservation



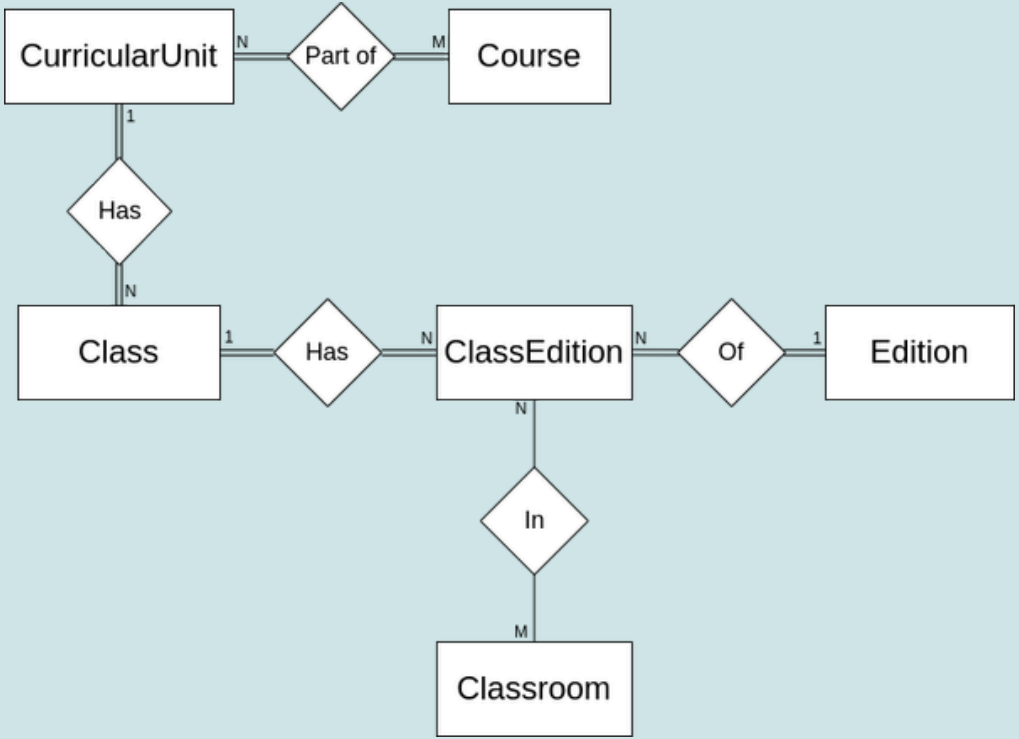
Accept Reservation



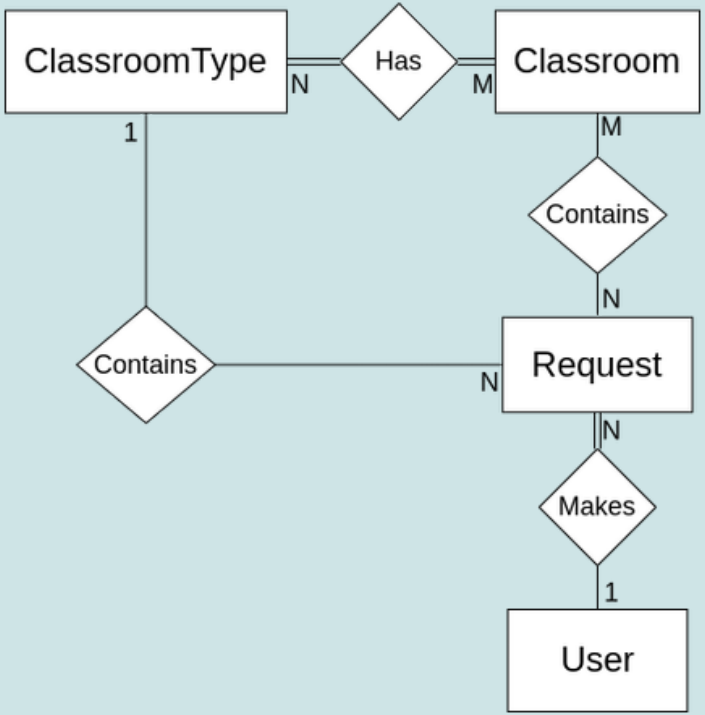
Deployment Diagram



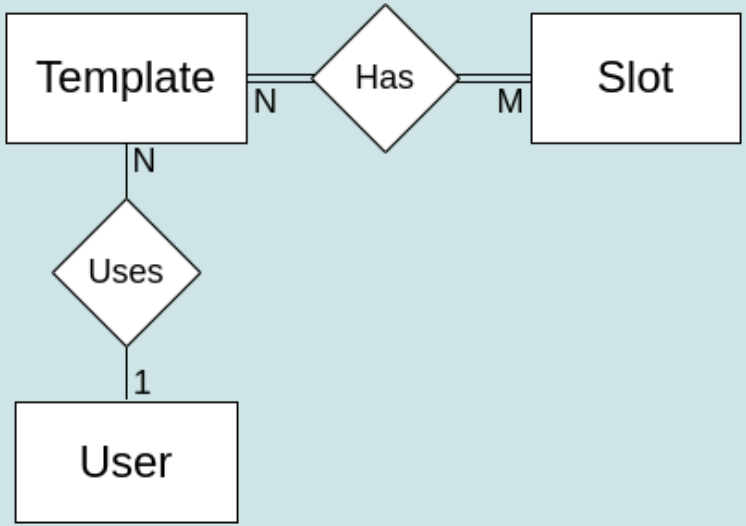
Schedule Section



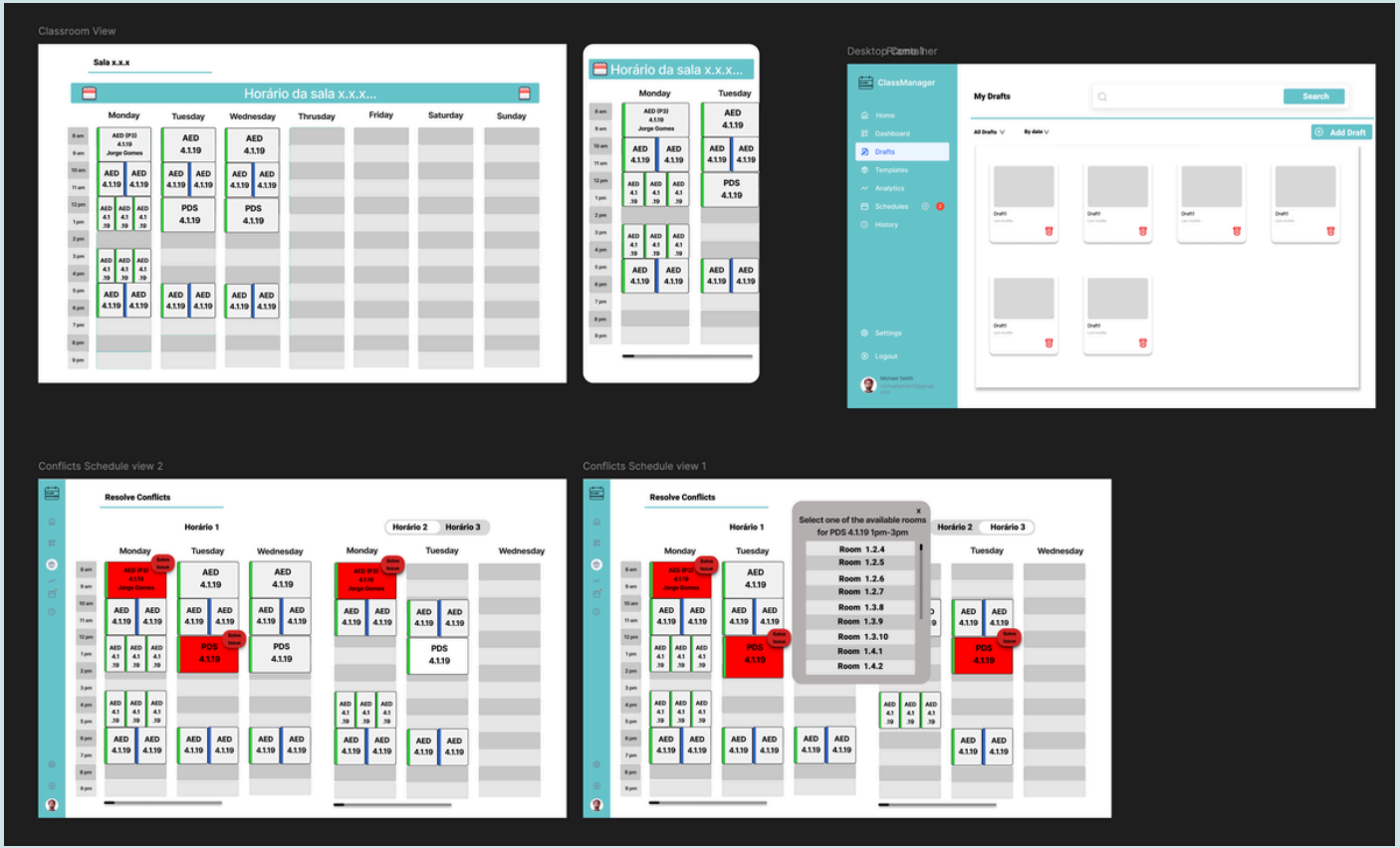
Classroom Requests Section



Templates Section

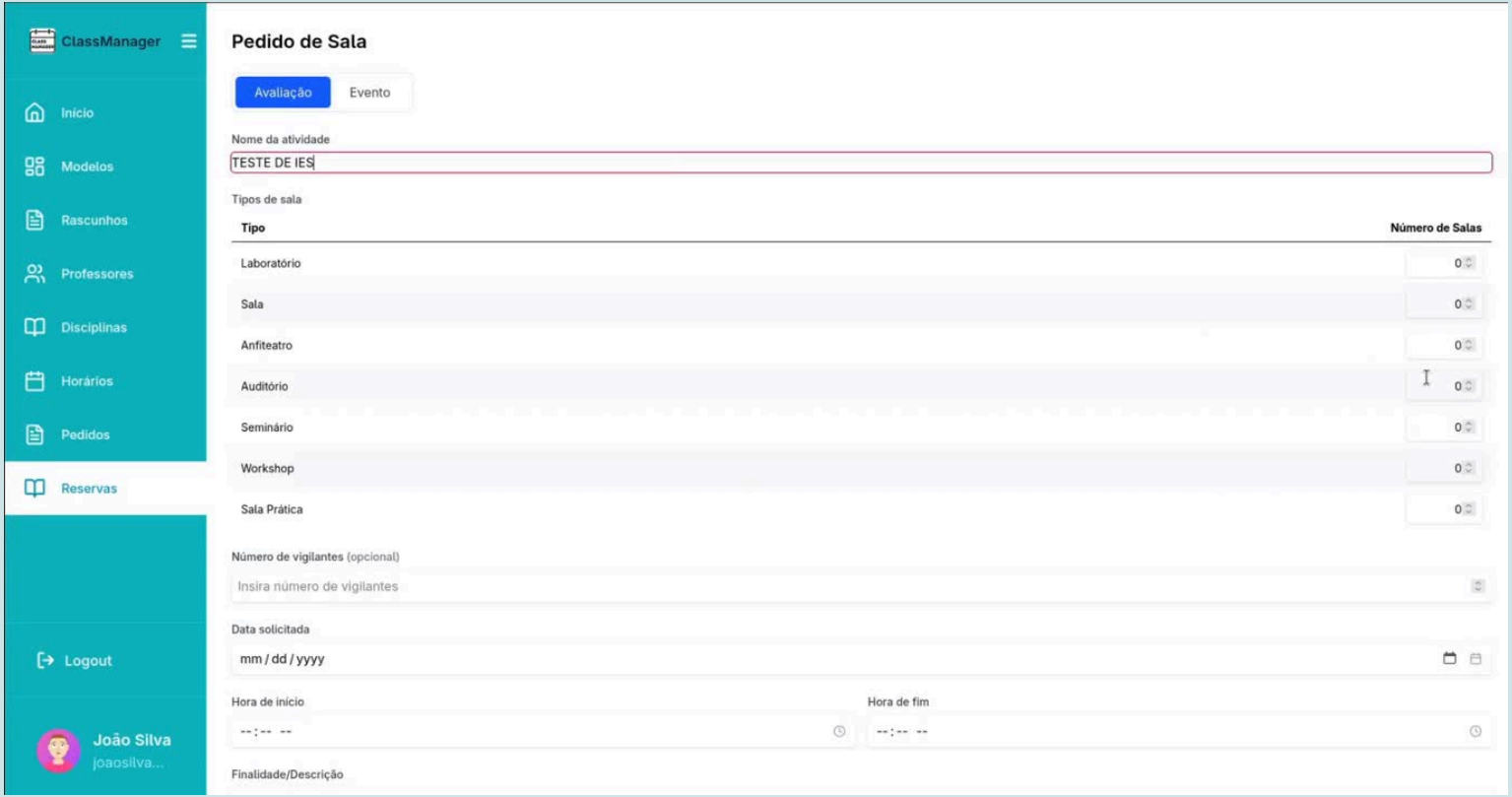


From Mockup to Solution



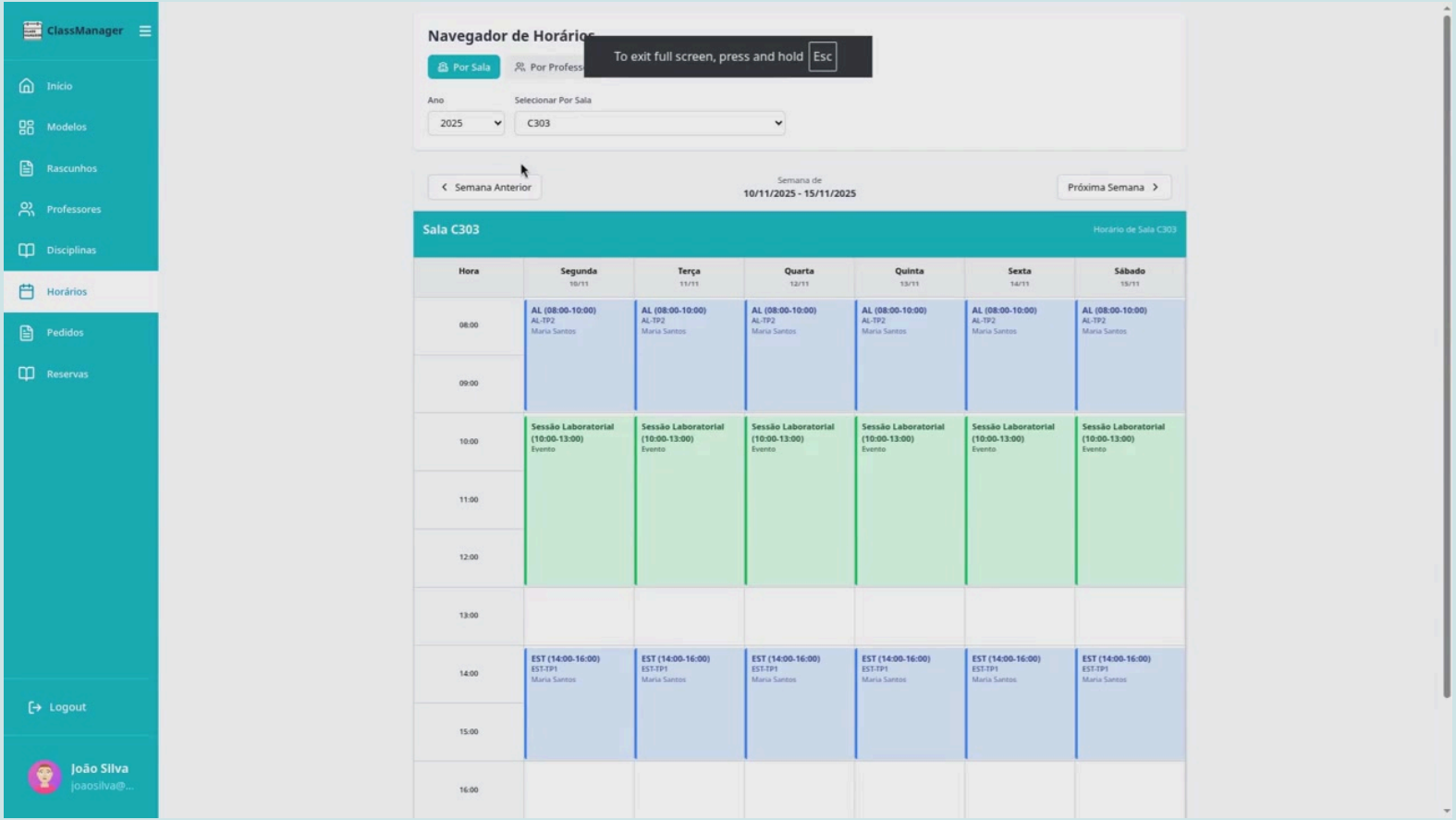
Booking a room for a test is often slow and uncertain due to administrative delays.

Book a Room in Seconds



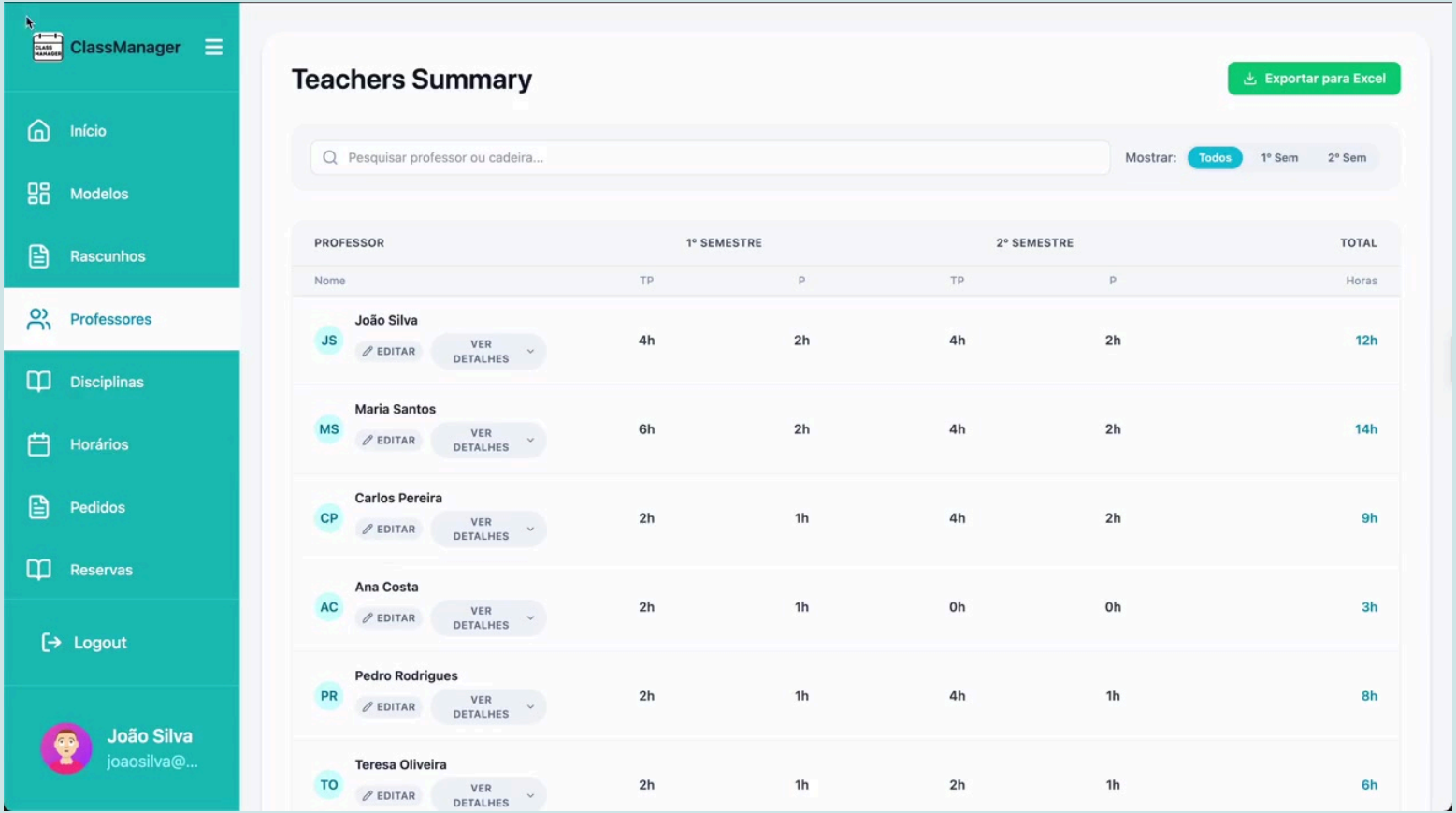
Finding a professor's current location or checking a room's availability requires juggling multiple documents and systems.

Visualize the schedules



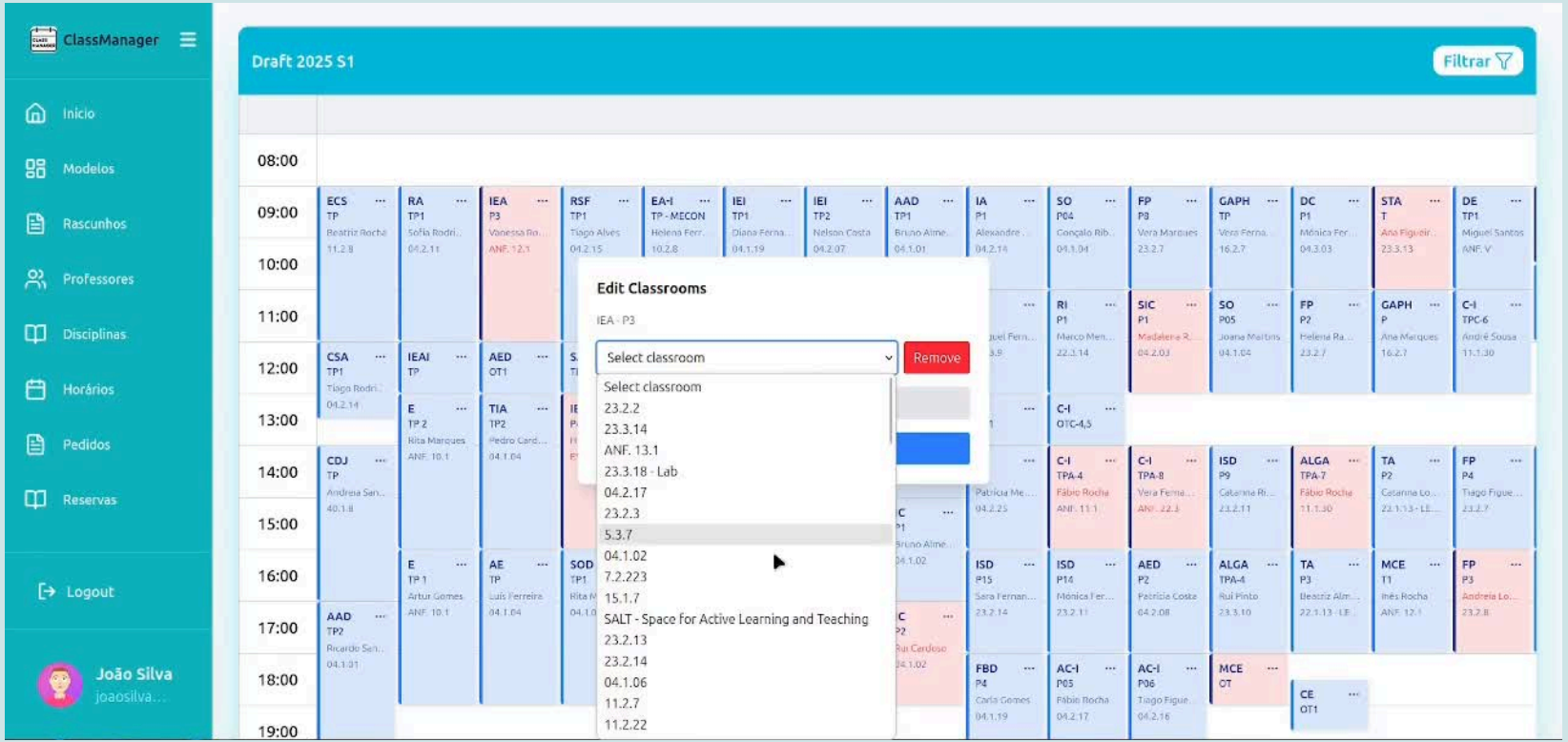
Changing a teacher's course load is a fragmented process.

Instant Updates



Optimising the schedule is complex, and changing the live schedule is risky.

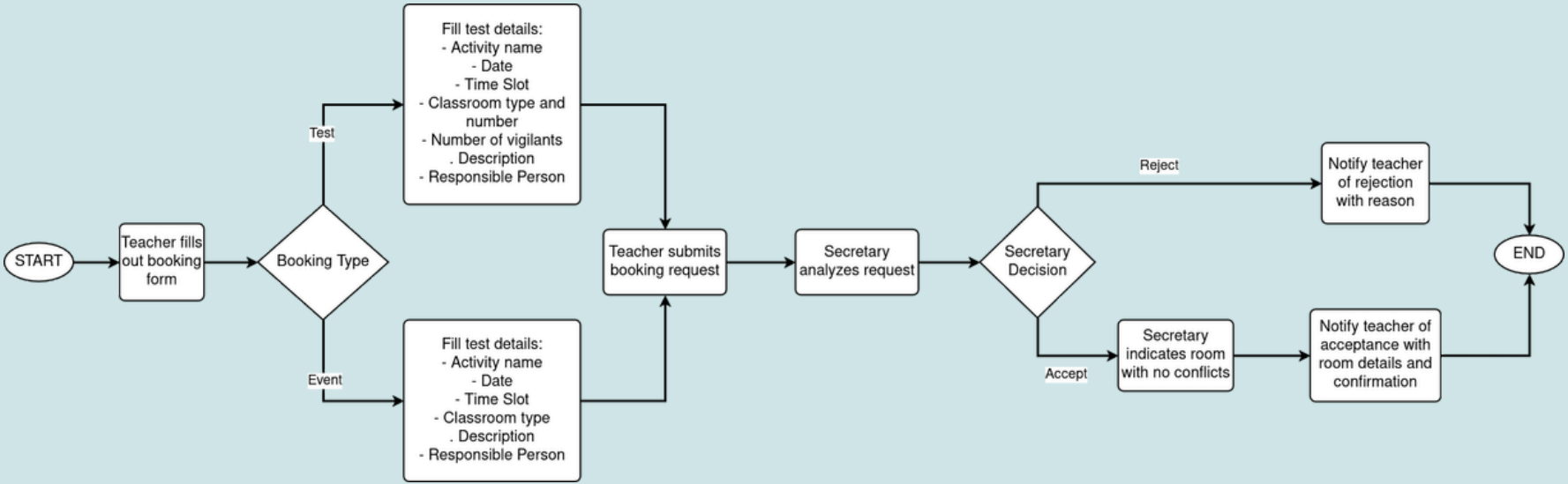
Visual Schedule Building



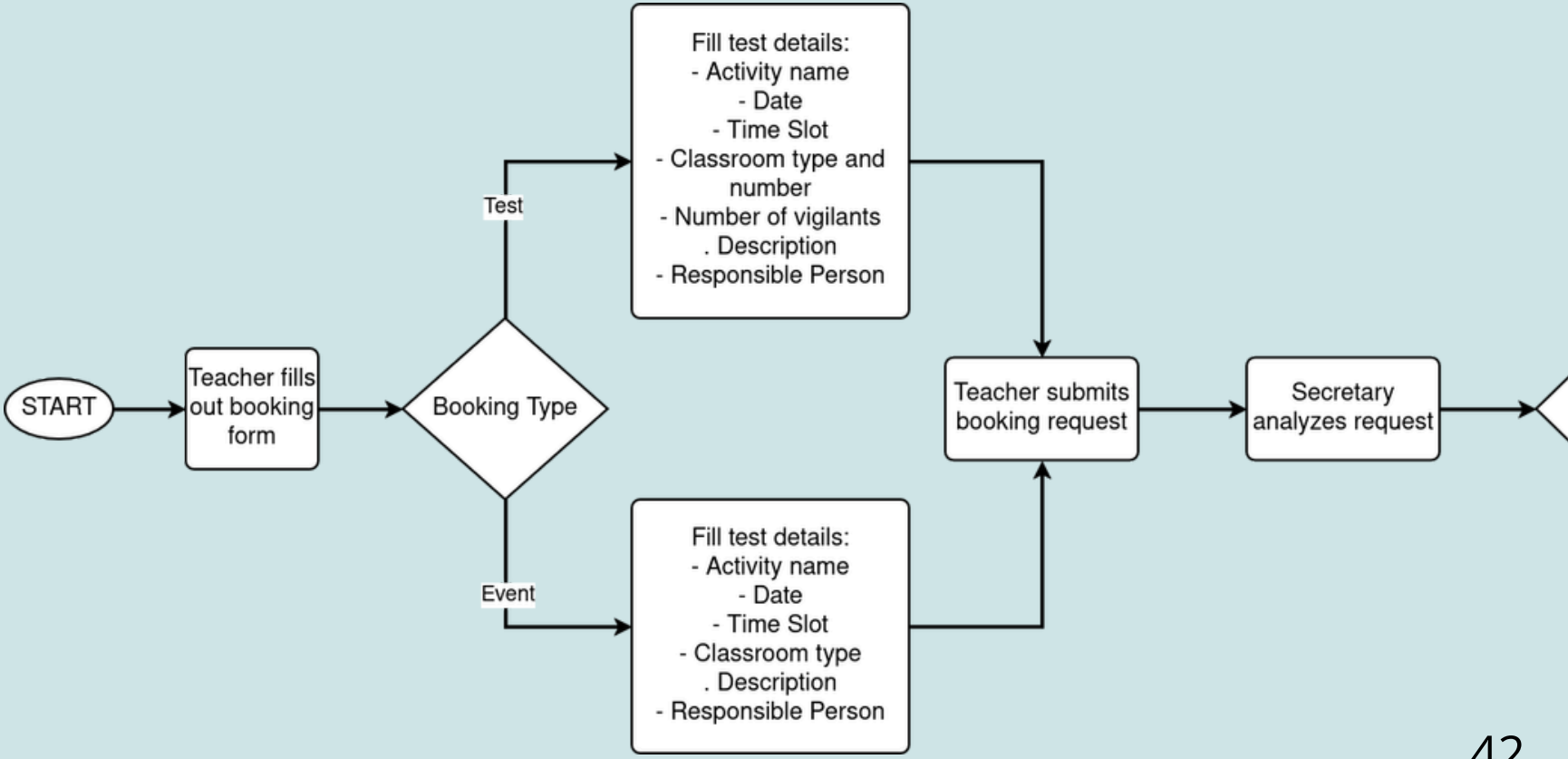
New Project Calendar

Modules	Date	Tasks List
Room Management	- to 09/12	1. Data Collection (11/11) 2. UI/UX Mockups (29/10 - 05/11) 3. Frontend Development (06/11 - 19/11) ← We are here 4. Database Implementation (20/11 - 26/11) 5. Backend for MVP terms (27/11 - 09/12)
Deployment and User Feedback	10/12 to 22/12	1. Deployment (10/12 - 12/12) 2. Qr Codes Placement (11/12 - 12/12) 3. Documentation (Instructions for User) (10/12 - 15/12) 4. User Feedback Analysis (13/12 - 22/12)

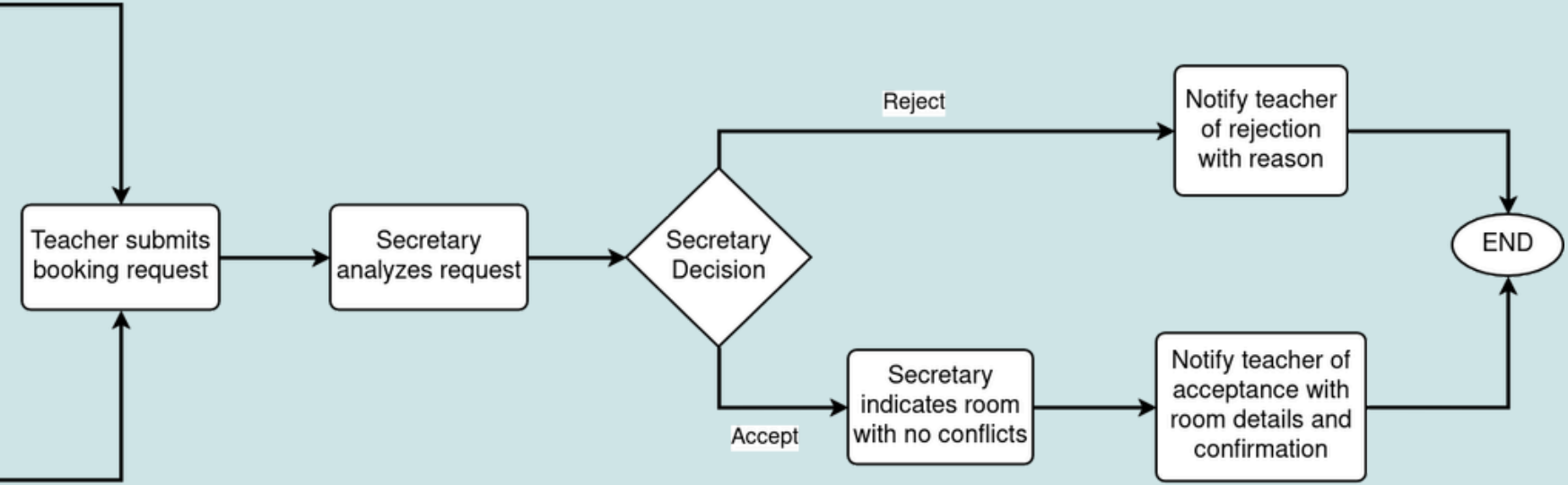
MVP - Room Management Flow Chart



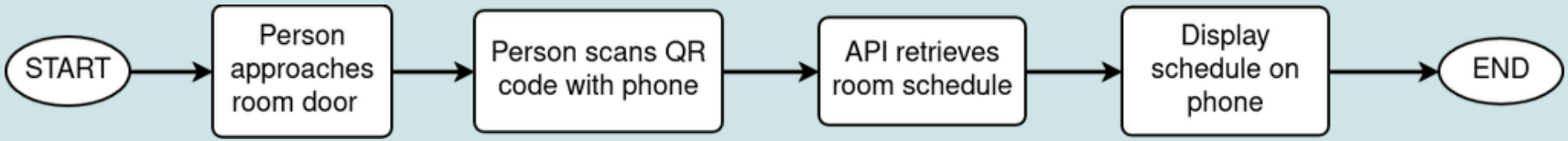
MVP - Room Management Flow Chart



MVP - Room Management Flow Chart



MVP - Room Management Flow Chart



Thank you!



Professor João Almeida





Daniel Ferreira

KEEP TRACK OF OUR WORK!



Margarida Ribeiro



José Bagagem



Simão Tojal



João Barreira



Gonçalo Almeida